



Bahir Dar University

Faculty Of Computing

Requirement Analysis Document (RAD) for Industrial project on **“A Smart Platform for Connecting Local Talent and On-Demand Services”**

Submitted to the faculty of computing in partial fulfillment of the requirements for
the degree of Bachelor of Science in Information Systems

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Declaration

We, Anwar Hussen, Mahlet Atnaфу, Mohammed Awol, and Mohammed Shiferaw, hereby declare that the Final Year Project entitled “**A Smart Platform for Connecting Local Talent and On-Demand Services**” is our original work and has been carried out under the guidance and supervision of our academic advisor.

This project has not been submitted, either wholly or in part, for the award of any degree, diploma, or other qualification at this or any other institution. All sources of information used in this project have been properly acknowledged in accordance with academic standards.

We further declare that this work complies with the rules and regulations of the university governing Final Year Project submissions.

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Roles and Responsibilities of the Group Members

List of Tasks		List of Members			
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Planning	Feasibility Study	✓	✓	✓	✓
	Resource Allocation	✓	✓	✓	✓
	Project Timeline	✓	✓	✓	✓
	Risk Identification	✓	✓	✓	✓
Requirements Gathering	Stakeholder Interviews & Surveys	✓	✓	✓	✓
	Business Process Mapping	✓	✓	✓	✓
	Functional vs. Non-Functional Listing	✓	✓	✓	✓
	Use Case Diagram development	✓	✓	✓	✓

Design	System Architecture Design	✓			
	Persistent Model Design				✓
	UI/UX Prototyping		✓	✓	
Implementation	Development Environment Setup	✓	✓	✓	✓
	Front-end & Back-end Coding	✓	✓	✓	✓
	Peer Code Reviews	✓	✓	✓	✓
	Version Control Management	✓	✓	✓	✓
Testing	Security & Penetration Testing	✓	✓	✓	✓
	Integration Testing	✓	✓	✓	✓
	User Acceptance Testing (UAT)	✓	✓	✓	✓
	Regression Testing	✓	✓	✓	✓

Table 1: Roles And Responsibilities Of The group Members

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List of Acronyms

Acronym	Full Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CRC	Class–Responsibility–Collaboration
CRUD	Create, Read, Update, Delete
DB	Database
DBMS	Database Management System
ERD	Entity Relationship Diagram
FR	Functional Requirement
HR	Human Resources
JWT	JSON Web Token
MVP	Minimum Viable Product
NFR	Non-Functional Requirement
OO	Object-Oriented
OOAD	Object-Oriented Analysis and Design
UC	Use Case
UI	User Interface
UML	Unified Modeling Language
UX	User Experience

Table 2: List of Acronyms

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Abstract

The rapid expansion of the digital economy has transformed global access to employment; however, in Ethiopia, significant barriers remain due to fragmented recruitment systems, a lack of professional verification, and the absence of integrated local payment methods. These inefficiencies, rooted in informal labor networks, lead to market mistrust, high unemployment, and a profound disconnect between skilled local talent and on-demand services. Current manual matching processes frequently result in wasted time and resources for both job seekers and recruiters.

The primary objective of this project is to develop SkillLink, a smart, multi-role digital platform designed to bridge the gap between informal service providers, job seekers, and clients. SkillLink merges the functionalities of an on-demand service marketplace and an intelligent job-matching system to create a unified digital ecosystem.

The project follows a structured system development lifecycle, utilizing UML modeling (including Use Case, Class, and Sequence diagrams) for design and a modular architecture for implementation. The methodology focuses on creating an AI-powered matching engine that connects talent based on specific skills, geographic location, and availability. Tools used include a frontend/backend stack supporting a responsive interface, localized for English and Amharic, and sandbox-based digital payment integrations.

SkillLink is developed as an academic Minimum Viable Product (MVP). The system architecture encompasses five distinct user roles—Customers, Service Providers, Job Seekers, Employers, and Administrators—supported by features such as real-time chat, interview support, and a transparent rating and review system.

The achieved outcome is a functional platform that demonstrates a transparent, end-to-end workflow—from AI-driven matching and verification to interview and payment. By meeting non-functional requirements for security (AES-256 encryption), performance ($\leq 2s$ response time), and scalability, the project validates the feasibility of a digital solution to foster economic inclusion and efficiency in the local labor market.

Chapter One: Introduction

1.1 Background

Technological advancement and increased internet penetration have fundamentally altered the landscape of professional engagement, making digital marketplaces essential for economic participation. In many regions, platforms that facilitate faster communication and efficient matching have become the backbone of the modern workforce. However, in Ethiopia, accessing verified, skilled service providers remains a persistent challenge due to a heavy reliance on informal labor networks and fragmented recruitment systems.

The Problem: Unemployment and Digital Barriers

In Ethiopia, unemployment rate is estimated at approximately 18.7%, with youth unemployment significantly higher at over 27% in urban centers. Despite the government's "Digital Ethiopia 2025" strategy successfully increasing internet users to 28.6 million (a 21.3% penetration rate), a massive "digital divide" and underemployment are critical socio-economic concerns. As of early 2025, the national unemployment usage gap" persists where 76% of the population living within network coverage does not yet use mobile internet for productive economic activities.

The primary barriers include a lack of professional trust and the absence of localized digital tools. Current global giants like Upwork and Fiverr primarily serve the international "gig economy," focusing on remote, cross-border digital tasks (e.g., software development or graphic design) paid in foreign currency. These platforms often neglect the local Ethiopian context, failing to support regional languages like Amharic, local payment gateways (e.g., Telebirr, M-Birr), or the on-demand physical services (e.g., plumbing, local tutoring) that dominate the domestic labor market. Without these features, many skilled local professionals remain invisible to the digital economy.

The Necessity for System Automation

Transitioning from manual, word-of-mouth recruitment to an automated system is no longer a luxury but a necessity for market efficiency. Automation through a platform like SkillLink addresses three critical pain points:

1. **Trust and Verification:** Manual systems lack accountability. Automation allows for systematic profile verification and a transparent rating system, reducing the risk of fraud.
2. **Matching Efficiency:** AI-driven matching can reduce the time-to-hire by up to 30-50% compared to traditional manual searching, ensuring that skills, location, and availability are perfectly aligned in real-time.
3. **Scalability:** Automated workflows enable the platform to handle thousands of concurrent users and transactions without the administrative bottlenecks inherent in informal networks.

The Proposed Solution: SkillLink

SkillLink is proposed as a comprehensive response to these challenges. It is a localized digital ecosystem designed to connect, verify, and intelligently match talent with clients in a single, trusted environment. By integrating AI-powered recommendations with real-time availability and secure, sandbox-based local payment systems, SkillLink aims to empower Ethiopian talent with digital visibility and provide employers with a streamlined, efficient, and transparent hiring process.

Based on the current socio-economic landscape of Ethiopia in 2025, the following section outlines the critical problems hindering the labor market and details how SkillLink provides specific, automated solutions.

1.2 Statement of the Problem

Despite the growing demand for digital hiring solutions in Ethiopia—where unemployment remains at approximately 18.7% and youth unemployment exceeds 27%—several core obstacles prevent the effective matching of talent and opportunity. Users frequently face difficulties in the following areas:

1. Trust and Verification Deficit

The informal nature of local transactions often lacks professional verification. With 90% of the Ethiopian workforce engaged in the informal economy, there is a pervasive lack of accountability, leading to unverified competition and reduced trust.

- **SkillLink Feature:** Rigorous Profile Verification & Admin Dashboard for user management and credential validation.

2. Inefficient and Costly Recruitment

Both businesses and individuals struggle to recruit skilled workers affordably. Relying on manual, word-of-mouth matching wastes significant time; for example, search and matching processes in traditional settings can take days or weeks.

- **SkillLink Feature:** AI-Based Recommendation Engine that automates matching based on skills, location, and availability to ensure response times of ≤ 3 seconds.

3. Lack of Digital Visibility and Portfolios

Local service providers often remain "invisible" to the digital economy. While mobile penetration has reached 63.8%, there are few platforms where workers can showcase a verified history of their professional experience or portfolios.

- **SkillLink Feature:** Profile Creation & Portfolio Management with a transparent rating and review system to build digital trust.

4. Integrated Communication Gaps

A lack of direct tools for messaging and interviewing creates friction. Delays in the hiring process are common when users must switch between multiple third-party apps to coordinate.

- **SkillLink Feature:** In-App Communication (Chat and Interview Support) including real-time scheduling and video calls.

5. Language and Financial Barriers

Many existing global platforms do not support Amharic or localized payment methods like Telebirr and CBE Birr, excluding a large portion of the population from participating in the digital market.

- **SkillLink Feature:** Multilingual Interface & Localized Digital Payment Integration (proposed via sandbox).

6. Inadequate Matching Technology

There is a lack of advanced filtering systems capable of handling end-to-end recruitment workflows within the local Ethiopian context.

- **SkillLink Feature:** UML-Modeled Modular Architecture designed specifically for five distinct roles (Customers, Providers, Job Seekers, Employers, and Admins) to handle the entire lifecycle of a service request.

Summary Table: Problem vs. SkillLink Solution

#	Identified Problem	SkillLink Solution/Feature
1	Market Mistrust	Profile Verification & Rating System
2	Recruitment Delays	AI-Powered Matching Engine
3	Worker Invisibility	Digital Portfolios & Profiles
4	Communication Friction	Integrated Chat & Interview Tools
5	Accessibility Barriers	Amharic Support & Local Payments
6	Manual Workflow	End-to-End Automated System

Table 3: Summary Table: Problem vs. SkillLink Solution

1.3 Objectives

1.3.1 General Objective

To design and develop SkillLink, an AI-powered digital ecosystem that automates the end-to-end process of talent matching, professional verification, and localized transactions to enhance recruitment efficiency and economic inclusion in Ethiopia.

1.3.2 Specific Objectives

To achieve the general objective, the following specific objectives have been defined, adhering to the SMART criteria:

- **System Architecture Design:** To design and implement a multi-role platform that supports four distinct user categories (Customer, Provider, Employer, and Administrator) with specialized workflows for each by the end of the development phase.
- **AI-Engine Development:** To develop an AI-powered matching engine that achieves a response time of < 3 seconds while connecting talent to requests based on skill sets, geographic proximity, and historical performance data.
- **Workflow Automation:** To integrate comprehensive modules for job posting, automated scheduling, and sandbox-based digital payment gateways to manage the full service-delivery lifecycle within the platform.
- **Real-Time Collaboration:** To facilitate direct in-app communication by implementing secure messaging and real-time video call features for live interviews and consultations.
- **Quality Assurance Framework:** To establish a trust-based ecosystem through mandatory profile verification protocols and a quantitative rating system designed to ensure 100% data integrity and accountability.
- **Administrative Oversight:** To develop a centralized dashboard that generates real-time analytics and reporting on platform usage, user verification status, and system performance metrics.
- **Localized Accessibility:** To provide a responsive, multilingual interface in English and Amharic, ensuring full functionality across mobile and desktop browsers (320px–1440px) to maximize local adoption.

1.4 Methodology

This project follows a **proposed, documentation-based methodology** since the current semester focuses on system analysis, design, and planning. The methodology explains how requirements are gathered, modeled, and how the system is planned to be designed and implemented in the next phase.

1.4.1 Requirement Gathering Methods

Requirement gathering was conducted using a combination of qualitative and analytical methods to ensure a clear understanding of user needs and system expectations. The following methods were used:

Interview

Informal and semi-structured interviews were conducted with potential users such as job seekers, service providers, employers, and recruiters. These interviews helped identify real-world challenges related to hiring, service booking, trust, and communication.

Questionnaire

Online questionnaires were prepared to collect broader input regarding preferred features, common problems in local hiring systems, and expectations from a digital talent platform.

Document Analysis

Existing platforms, job portals, and service marketplaces were reviewed to analyze common workflows, features, and system limitations.

Literature Review

Academic papers, articles, and case studies related to AI-based matching systems, on-demand platforms, and recruitment systems were reviewed to identify best practices and design standards.

After collecting the requirements, they were structured into **functional and non-functional requirements**. The system requirements are modeled using an **Object-Oriented (OO) approach**, as it supports modularity, scalability, and reuse, which are essential for a multi-role platform like SkillLink.

Tools used for requirement modeling: - UML diagrams (Use Case, Class, Sequence diagrams)
- Draw.io / StarUML for visual modeling - Documentation tools (MS Word / Google Docs)

1.4.2 Analysis and Design Methodology

The project adopts an Object-Oriented Analysis and Design (OOAD) methodology. This approach was selected because it effectively models real-world entities—such as users, jobs, services, and transactions—as objects with well-defined attributes and behaviors, making the system easier to analyze, design, and extend.

During the analysis and design phase, the following activities are performed:

- Identification of system actors and functional requirements
- Modeling system interactions and workflows
- Designing system structure, behavior, and overall architecture

To achieve this, the system is modeled using the following specific UML diagrams:

- Use Case Diagram – to identify system actors and their interactions with the system
- Class Diagram – to represent system classes, attributes, methods, and relationships
- Sequence Diagram – to illustrate the interaction and message flow between objects over time
- Activity Diagram – to model the workflow of key system processes
- State Chart Diagram – to show the state transitions of jobs or service requests

UML modeling tools such as StarUML and Draw.io are used to design and document these diagrams in a standardized and structured manner.

1.4.3 Implementation Methodology

Although the actual implementation will be done in the next semester, the following proposed implementation methodology and tools are identified:

- **Programming Languages:** JavaScript / TypeScript
- **Frontend Framework:** Next.js with Tailwind CSS

- **Backend Framework:** Node.js with Express or Next.js API routes
- **Database Management System:** PostgreSQL or MongoDB
- **Authentication:** JWT-based authentication
- **AI Module:** Python-based machine learning models or pre-trained AI services
- **Application Server:** Node.js runtime environment
- **Version Control:** Git and GitHub

Agile development with iterative sprints is proposed for building, testing, and refining system modules.

1.5 Feasibility Study

Feasibility Study

This section evaluates the feasibility of developing A Smart Platform for Connecting Local Talent and On-Demand Services from technical, economic, operational, and time perspectives.

1.5.1 Technical Feasibility

The project is technically feasible as it utilizes proven, industry-standard technologies for web and mobile application development. The system architecture follows a modular approach, incorporating a frontend interface, backend services, database management, and AI-based matching mechanisms. By leveraging pre-trained AI models and established third-party APIs for communication and payments, the technical complexity and initial development time are significantly reduced.

Despite this feasibility, the following technical risks and mitigation strategies are identified:

Technical Risk	Impact	Mitigation Strategy
Data Security & Privacy	High	Implementing AES-256 encryption, TLS 1.2+ protocols, and JWT-based authentication to protect sensitive user data.
Integration Failure	Medium	Utilizing sandbox environments for payment gateways (e.g., Telebirr, CBE Birr) to ensure stability before full integration.
Scalability Bottlenecks	Medium	Adopting a modular architecture and cloud-based hosting to handle potential spikes in concurrent user traffic.
Matching Accuracy	Low	Continuously refining the AI-matching engine through iterative testing and user feedback loops to improve skill-location alignment.

Table 4: Feasibility technical risks and mitigation strategies

1.5.2 Economic Feasibility

The project is economically feasible because it focuses on developing a Minimum Viable Product (MVP) by leveraging cost-effective, open-source technologies. By utilizing free frameworks such as React, Node.js, or Flutter and open-source database solutions like PostgreSQL or MongoDB, the project eliminates the need for expensive proprietary software licenses. Furthermore, the use of sandbox environments for local payment integrations, such as Telebirr and CBE Birr, allows for comprehensive testing of financial workflows without incurring transaction costs or requiring immediate commercial banking agreements.

In terms of specific cost estimates, the primary expenses are limited to infrastructure and identity. Domain registration for a .et or .com extension typically ranges from 800 to 1,500 ETB annually, while entry-level cloud hosting for an MVP can be managed for approximately 1,500 to 4,000 ETB per month, or even less if utilizing free-tier credits from providers like AWS or Google Cloud. A small contingency fund of roughly 5,000 ETB is sufficient to cover any additional third-party API usage or data acquisition needs during the development phase.

In the long term, the platform is designed for financial sustainability. It can transition from an academic prototype into a revenue-generating business by charging service commissions (e.g., 5-10% per transaction), offering premium employer listings for higher visibility, or implementing subscription-based tiers for professional service providers. Given that Ethiopia's digital economy is projected to grow significantly by 2025, these monetization strategies provide a clear path for the platform to recover initial costs and support long-term maintenance.

1.5.3 Operational Feasibility

The platform is designed to be user-friendly and accessible to multiple user roles, including customers, service providers, job seekers, employers, and administrators. Features such as profile verification, ratings, multilingual support (English and Amharic), and an administrative control panel ensure smooth operation and trust among users.

1.5.4 Time Feasibility

The project is highly feasible within the allocated academic timeframe by adopting an Agile development methodology. This iterative approach breaks the project into manageable two-week sprints, allowing for parallel development of the frontend, backend, and AI matching modules. By focusing on a Minimum Viable Product (MVP), the team ensures that core functionalities are completed and validated before the final deadline.

Project Timeline and Milestones

The development schedule is structured around a 12-week timeline, with major milestones clearly defined to ensure a systematic progression:

1. Phase 1: Requirements & System Design (Weeks 1–3)

- Conducting needs assessment and finalizing system specifications.
- Developing UML diagrams (Use Case, Class, Sequence, and Activity diagrams) to serve as the development blueprint.
- Milestone: Completion of the Software Requirements Specification (SRS) and System Design Document.

2. **Phase 2:** Sprint 1 – Foundation & Authentication (Weeks 4–5)
 - Setting up the database schema and cloud environment.
 - Implementing multi-role user registration and JWT-based authentication.
 - Milestone: Functional user onboarding and dashboard shell.
3. **Phase 3:** Sprint 2 – Core Logic & AI Matching (Weeks 6–7)
 - Developing the service booking and job posting modules.
 - Implementing the AI matching engine for skill-based recommendations.
 - Milestone: Integrated smart-matching functionality between Providers and Clients.
4. **Phase 4:** Sprint 3 – Communication & Payments (Weeks 8–9)
 - Integrating in-app messaging and real-time video call features for interviews.
 - Setting up sandbox-based payment gateways for local currency transactions.
 - Milestone: End-to-end service delivery workflow from booking to payment.
5. **Phase 5:** Testing, Evaluation & Final Documentation (Weeks 10–12)
 - Conducting system testing (unit, integration, and user acceptance testing).
 - Performing final bug fixes and performance optimization (response time ≤ 2s).
 - Completing the final project report and defense preparation.
 - Milestone: Final MVP launch and project delivery.

To mitigate time-related risks, the team utilizes a priority-based backlog. This ensures that "must-have" features (e.g., registration, matching, and payments) are finalized early, while "nice-to-have" features are only implemented if time permits. This structure guarantees a functional, high-quality system within the academic constraints.

1.6 Beneficiaries / Significance of the Project

The SkillLink platform benefits multiple stakeholders directly and indirectly by improving access, efficiency, and trust in the local talent and on-demand service market.

Direct Beneficiaries

Job Seekers and Service Providers

- Gain increased digital visibility through structured and verified profiles
- Access fair and transparent job and service opportunities
- Receive AI-based recommendations that match skills, location, and availability
- Build professional credibility through ratings, reviews, and work history

Employers and Customers

- Save time and operational cost through intelligent talent and service matching
- Access verified and qualified candidates and service providers
- Reduce hiring and service delivery risks through transparent reviews and profiles
- Benefit from streamlined communication and secure digital transactions

System Administrators

- Monitor platform activity through a centralized administrative dashboard
- Manage user verification, dispute resolution, and system compliance efficiently
- Gain analytical insights to support system improvement and decision-making

Indirect Beneficiaries

Local Communities

- Increased access to employment and service opportunities
- Reduction of informal and unreliable hiring practices

- Promotion of trust-based digital service interactions

Local Economy

- Improved workforce utilization and productivity
- Encouragement of digital participation in economic activities
- Support for small-scale service providers and freelancers to generate income

Academic and Research Community

- Serves as a practical case study for AI-driven recruitment and service platforms
- Contributes to research in information systems, system modeling, and digital marketplaces
- Provides reference material for future academic projects and system development

Technology and Innovation Ecosystem

- Encourages adoption of modern system analysis and design methodologies
- Supports digital innovation aligned with local market needs
- Overall, the project contributes to the digital transformation of local labor markets by promoting efficiency, transparency, inclusivity, and trust among all stakeholders.

1.7 Limitations of the Project

This project, entitled “A Smart Platform for Connecting Local Talent and On-Demand Services,” is designed to address the challenges of connecting service providers, job seekers, employers, and customers through a unified digital platform. Within the scope of this project, the system partially solves the organizational and market-level problems related to inefficient recruitment, lack of trust, manual matching processes, and limited visibility of local talent.

1.7.1 Scope and Extent of Problem Solving

The system provides a centralized solution that enables users to register, create verified profiles, post jobs or services, search and match talent using intelligent filtering mechanisms, communicate through in-app messaging and video calls, and complete transactions using integrated digital payment methods. To this extent, the platform significantly improves the efficiency, transparency, and reliability of local talent recruitment and on-demand service delivery.

However, the project focuses on delivering a Minimum Viable Product (MVP) within the given academic timeframe. As a result, the system demonstrates core functionalities rather than a fully scaled commercial solution. While the platform addresses the major problems of manual recruitment, unverified providers, and inefficient matching, it does not fully eliminate all operational, regulatory, and scalability challenges faced in real-world deployment.

1.7.2 Services and Deliverables Achieved

Within the available time and resources, the project successfully delivers the following core services:

- User registration and role-based access for customers, service providers, job seekers, employers, and administrators
- Profile creation with basic verification mechanisms
- Job posting and service listing functionalities
- Intelligent matching based on skills, location, and availability
- In-app communication including messaging and basic video interview support
- Secure transaction handling using test or sandbox payment integrations
- Rating and review system to enhance trust and accountability
- Administrative dashboard for monitoring users and system activities

These deliverables meet the essential functional requirements of the proposed system and validate the feasibility of the solution.

1.7.3 Challenges and Constraints Encountered

Several challenges and constraints were encountered during the development of the project, which limited the full realization of the original plan:

- **Limited time frame:** which restricted the depth of feature implementation and extensive testing
- **Insufficient real-world data:** for training and evaluating advanced AI-based matching models
- **Technical complexity:** associated with integrating real-time video communication and payment systems
- **Internet connectivity limitations:** which affect real-time features and system reliability in some environments
- **Resource constraints:** including limited access to paid APIs, cloud infrastructure, and production-level services

These issues influenced both development speed and feature completeness.

1.7.4 Activities Left Undone Due to Constraints

- Due to the above challenges and constraints, the following planned activities were not fully implemented or were partially completed:
- Deployment of the system in a live production environment with real users
- Advanced AI model training using large-scale, real-world datasets
- Support for multiple international and local payment gateways beyond test modes
- Comprehensive security auditing and penetration testing

- Full scalability optimization for high concurrent user traffic
- Extensive user acceptance testing and long-term performance evaluation

Despite these limitations, the project meets the academic standards expected for a final-year project and provides a strong foundation for future enhancement, scalability, and real-world deployment.

1.8 Scope of the Project

The scope of this project focuses on the analysis, design, and documentation of SkillLink: A Smart Platform for Connecting Local Talent and On-Demand Services. The system is developed as an academic Minimum Viable Product (MVP) and defines the functional boundaries of the proposed solution.

In-Scope

The project includes the following system functionalities and components:

- User registration and authentication
- Role-based access for Customers, Service Providers, Job Seekers, Employers, and Administrators
- Profile creation and management, including basic verification
- Job posting and service request creation
- AI-based recommendation and matching based on skills, location, and availability
- In-app communication (chat and interview support)
- Digital payment integration (proposed and sandbox-based)
- Rating and review system to promote trust and accountability
- Administrative dashboard for user management, verification, and reporting
- System modeling using UML diagrams (Use Case, Class, Sequence, Activity, State Chart, and Architecture diagrams)

- Documentation of functional and non-functional requirements

Out-of-Scope

The following features are explicitly excluded from the scope of this project:

- Full-scale commercial deployment and production hosting
- Legal enforcement of contracts or employment agreements
- Government-level identity or credential verification integrations
- Offline system operation without internet connectivity
- Advanced AI model training using large-scale real-world datasets
- Multi-country or international labor market support
- Banking services or financial institution functionality
- Long-term system maintenance and post-deployment support

Scope Boundary

The project is limited to demonstrating the feasibility, structure, and design of the proposed system within the constraints of academic time and resources. While core workflows are modeled and validated, the system is not intended to function as a fully deployed commercial platform.

1.9 Organization of the Project

This project documentation is organized into five main chapters as described below:

1. Chapter One: Introduction

Presents the background of the study, problem statement, objectives, methodology, beneficiaries and significance of the project, scope, and organization of the document.

2. Chapter Two: System Features

Describes the existing system and the proposed system. It also presents system features, functional and non-functional requirements, business rules, and system models such as use case diagrams, state chart diagrams, activity diagrams, and sequence diagrams.

3. Chapter Three: System Design

Focuses on the detailed design of the system. This chapter includes architectural design, component modeling, deployment modeling, detailed design class models, persistent data models, user interface design, and access control and security mechanisms.

4. Chapter Four: Implementation

Describes the technologies, tools, and approaches used to implement the proposed system, including frontend, backend, database, and integration components.

5. Chapter Five: Testing and Evaluation

Presents the testing strategies, test cases, evaluation criteria, and analysis of system performance and functionality.

Each chapter is structured to provide a logical progression from system requirements to design, implementation, and evaluation, ensuring a clear understanding of the proposed system.

Chapter 2

2.1 The Existing System

The existing system for connecting local talent and on-demand services in Ethiopia is largely informal, fragmented, and inefficient. Most job seekers, service providers, employers, and customers rely on personal networks, word-of-mouth referrals, social media, or intermediary agents to find work or services. Common tools and platforms used include:

- WhatsApp – for direct messaging and group coordination
- Facebook – for posting service offers, job requests, and networking
- Telegram – for community announcements and local group discussions
- Phone Calls and SMS – for personal communication and confirmations
- Physical Notice Boards – in local markets, institutions, or workplaces
- Informal Agents / Intermediaries – individuals connecting service providers and customers

These approaches lack structure, transparency, and accountability.

Currently, there is no centralized digital platform that integrates job recruitment and service booking while ensuring professional verification. Employers and customers often struggle to identify skilled and trustworthy individuals, while service providers and job seekers face limited visibility and inconsistent access to opportunities. The absence of standardized profiles, skill validation mechanisms, and performance tracking makes it difficult to assess competence and reliability.

Communication in the existing system is manual and disconnected, typically conducted through phone calls or third-party messaging applications like WhatsApp and Telegram. This leads to delays, miscommunication, and scheduling conflicts. Additionally, payments are mostly handled in cash or informal transfers, increasing the risk of disputes, fraud, and lack of financial traceability.

Overall, the existing system is characterized by:

- Manual and time-consuming recruitment and service matching processes
- Lack of professional verification and trust mechanisms
- Limited digital visibility for local talent
- Absence of integrated communication and payment systems
- No structured data for tracking performance, history, or analytics

These limitations significantly reduce efficiency, reliability, and trust in the local labor and service market.

2.2 Proposed System

The proposed system, **SkillLink: A Smart Platform for Connecting Local Talent and On-Demand Services**, introduces a unified, intelligent, and secure digital solution that overcomes the shortcomings of the existing system. SkillLink is designed as a multi-role platform that connects customers, service providers, job seekers, employers, and administrators within a single ecosystem.

The system leverages AI-powered matching to recommend suitable talent based on skills, location, availability, and historical performance. Unlike the existing manual process, SkillLink automates recruitment and service booking workflows, significantly reducing time and effort for all stakeholders.

SkillLink integrates essential features such as verified user profiles, portfolio uploads, real-time chat and video communication, secure digital payments, and transparent rating and review mechanisms. The platform supports localized payment methods and multilingual interfaces (English and Amharic), ensuring accessibility and inclusivity for local users.

Key improvements introduced by the proposed system include:

- Centralized digital marketplace for jobs and services
- Intelligent matching using AI-based recommendations
- Secure authentication and role-based access control
- Integrated communication tools (chat and video calls)
- Digital payment handling with transaction tracking
- Trust-building through verification, ratings, and reviews
- Administrative oversight via dashboards and analytics

By providing an end-to-end workflow—from posting and matching to payment and feedback—the proposed system creates a reliable, efficient, and scalable digital environment for talent and service exchange.

2.3 Requirement Analysis

Requirement analysis identifies and documents the functional needs of “**A Smart Platform for Connecting Local Talent and On-Demand Services.**” This process ensures that the system meets user expectations and addresses the challenges of inefficient recruitment, lack of trust, and poor coordination between service providers, job seekers, employers, and customers.

The requirements are derived from the project objectives, problem statement, and expected outcomes, and they form the basis for system design, implementation, and evaluation.

2.3.1 Functional Requirements

The functional requirements define the services and operations the system must perform. Each requirement is uniquely identified and assigned a priority level.

Functional Requirements List

Requirement ID	Requirement Description	Priority
FR-1	The system shall allow users to register and log in securely.	High
FR-2	The system shall support multiple user roles (Customer, Service Provider, Job Seeker, Employer, Admin).	High
FR-3	The system shall allow users to create, update, and manage personal profiles.	High
FR-4	The system shall allow service providers and job seekers to upload skills, portfolios, and credentials.	High
FR-5	The system shall provide profile verification to enhance trust among users.	Medium
FR-6	The system shall allow employers to post job vacancies.	High
FR-7	The system shall allow customers to post service requests and bookings.	High
FR-8	The system shall recommend suitable service providers or job candidates based on skills, location, and availability.	High

FR-9	The system shall allow users to communicate via in-app messaging.	High
FR-10	The system shall support video communication for interviews or service discussions.	Medium
FR-11	The system shall allow users to make and receive payments using integrated digital payment methods.	High
FR-12	The system shall record and display transaction history.	Medium
FR-13	The system shall allow users to rate and review completed jobs or services.	Medium
FR-14	The system shall provide an administrative dashboard for monitoring users and activities.	High
FR-15	The system shall allow administrators to manage users, resolve disputes, and generate reports.	High

Table 5: Functional Requirements List

2.3.2 System Use Case

System use cases describe how different actors interact with the system to achieve specific goals. These use cases are derived directly from the functional requirements and are governed by specific business rules to ensure security, efficiency, and fairness.

Actors

- **Customer:** Individuals seeking on-demand services (e.g., plumbing, tutoring).
- **Service Provider:** Individuals offering specialized skills for short-term tasks.
- **Job Seeker:** Individuals applying for long-term professional roles.
- **Employer:** Organizations or recruiters seeking talent for long-term employment.
- **Administrator:** System overseers responsible for verification and platform health.

2.3.2.1 Use Case Diagram

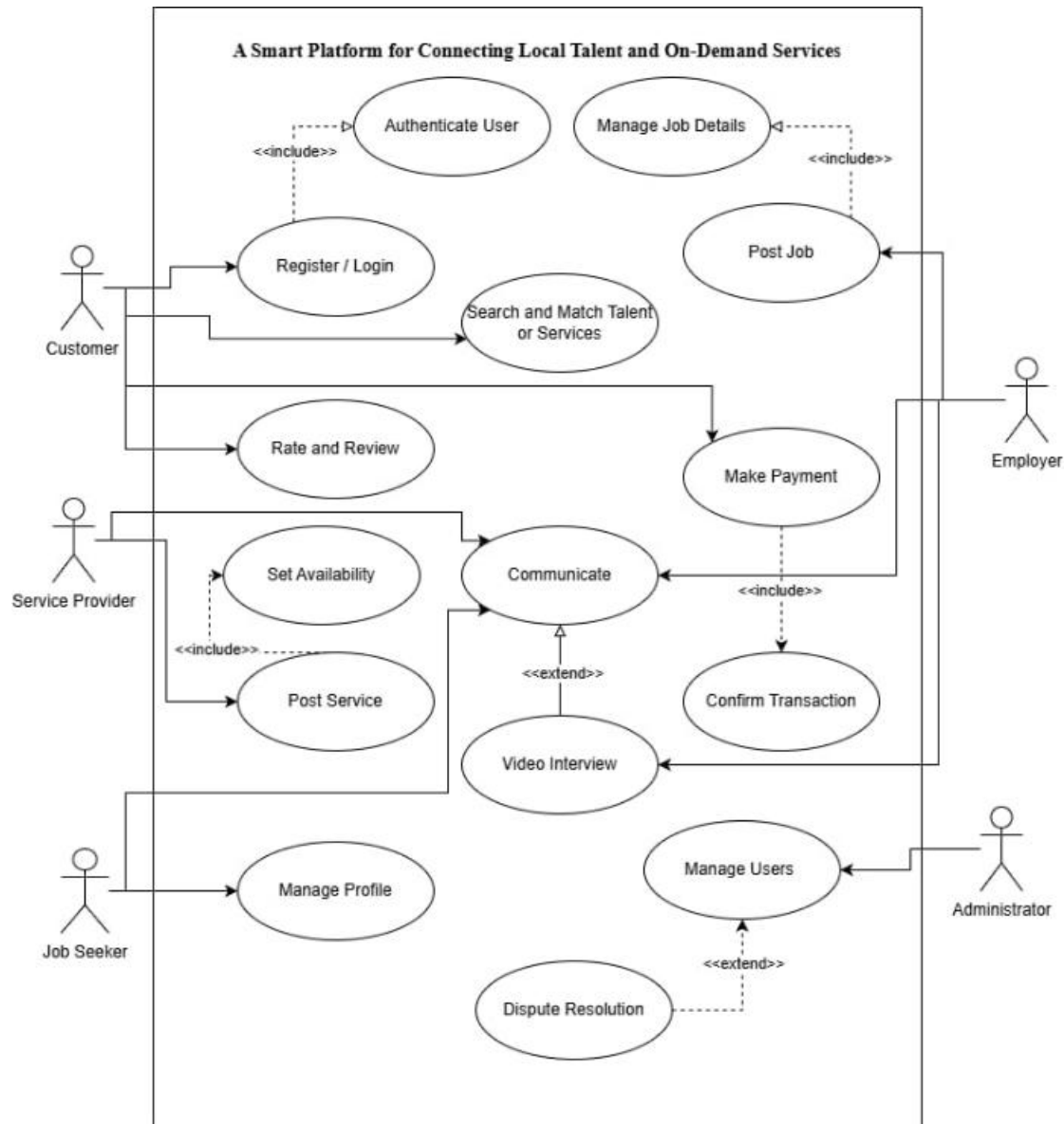


Figure 1: Illustrates the use case diagram of the system, showing interactions between actors and system functionalities, including include, extend, and inheritance relationships.

2.3.2.2 Use Case Documentation

The use case diagram visually represents the system's functionality and interactions between actors and the system. To ensure a comprehensive understanding of the SkillLink platform, the following detailed documentation covers all key use cases identified in Figure 2.1.

Use Case 1: User Registration and Login

- Use Case ID: UC-01
- Actor: All Users (Customer, Service Provider, Job Seeker, Employer, Administrator)
- Description: Allows users to create a role-specific account and securely access the platform.
- Pre-condition: User has an internet connection and access to the SkillLink interface.
- Post-condition: User is authenticated and redirected to their respective dashboard.
- Priority: High
- Basic Course of Action:
 1. User selects the "Register" or "Login" option.
 2. User enters required credentials (Email/Phone, Password, and Role).
 3. System validates input and includes Authenticate User workflow.
 4. System grants access and displays the dashboard.
- Alternate Course of Action:
 1. Invalid credentials → System displays an error message.
 2. User forgets password → System initiates the password recovery module.

Use Case 2: Manage Profile

- Use Case ID: UC-02
- Actor: All Users
- Description: Allows users to update personal information, professional skills, and portfolio documents.
- Pre-condition: User is logged into the system.
- Post-condition: Profile information is updated and visible to relevant actors.
- Priority: High
- Basic Course of Action:
 1. User navigates to "Profile Settings."

2. User updates details (Bio, Skills, Experience, or Portfolio).
3. System validates data formats.
4. System updates the persistent data model.

Use Case 3: Post Job / Post Service

- Use Case ID: UC-03
- Actor: Employer, Service Provider
- Description: Allows employers to post vacancies and providers to list on-demand services.
- Pre-condition: User is logged in with a completed profile.
- Post-condition: The post is published and added to the matching pool.
- Priority: High
- Basic Course of Action:
 1. User selects "Post Job" or "Post Service."
 2. User enters details (Description, Category, Location, Budget/Rate).
 3. System Includes Manage Job Details or Set Availability.
 4. System validates and publishes the post.

Use Case 4: Search and Match Talent or Services

- Use Case ID: UC-04
- Actor: Customer, Employer, Job Seeker, Service Provider
- Description: The AI engine identifies and presents optimal connections based on skills and location.
- Pre-condition: A job or service must be active in the system.
- Post-condition: A list of recommended matches is presented to the user.
- Priority: High
- Business Rule (Bias Prevention): The AI matching logic utilizes Attribute Masking to ensure rankings are based strictly on merit, excluding gender, age, or ethnicity from the decision-making process.
- Basic Course of Action:
 1. User initiates a search or the system triggers an automated recommendation.
 2. AI Engine filters data based on skill-alignment and geographic proximity.
 3. System displays the ranked results.

Use Case 5: Communicate (Chat / Video Call)

- Use Case ID: UC-05
- Actor: Customer, Service Provider, Job Seeker, Employer
- Description: Facilitates real-time interaction between parties to discuss requirements or conduct interviews.
- Pre-condition: A match has been accepted or a query initiated.
- Post-condition: Communication history is logged.
- Priority: Medium
- Basic Course of Action:
 1. User opens the communication module with a specific match.
 2. User sends text-based messages via Chat.
 3. Extend: User may initiate a Video Interview for real-time live screening.

Use Case 6: Make Payment

- Use Case ID: UC-06
- Actor: Customer, Employer
- Description: Handles the financial transaction for services rendered or recruitment fees.
- Pre-condition: A service or job contract is marked for payment.
- Post-condition: Transaction is processed, and a receipt is generated.
- Priority: High
- Basic Course of Action:
 1. User selects "Pay Now."
 2. System includes the Confirm Transaction workflow via sandbox integration.
 3. Payment is processed through local gateways (e.g., Telebirr, CBE Birr).

Use Case 7: Rate and Review

- Use Case ID: UC-07
- Actor: Customer, Service Provider
- Description: Allows parties to provide feedback and build platform trust.
- Pre-condition: A transaction or service is completed.

- Post-condition: Ratings are updated on the target's public profile.
- Priority: Medium
- Basic Course of Action:
 1. User selects "Rate Service."
 2. User provides a numerical rating and a text review.
 3. System calculates the new average rating.

Use Case 8: Manage Users (Admin)

- Use Case ID: UC-08
- Actor: Administrator
- Description: Allows the admin to oversee platform operations, verify users, and handle disputes.
- Pre-condition: Administrator is logged into the Admin Dashboard.
- Post-condition: System integrity is maintained.
- Priority: High
- Basic Course of Action:
 1. Admin monitors user activities and verification requests.
 2. Admin validates credentials or suspends accounts.
 3. Extend: Admin initiates Dispute Resolution if a conflict is flagged between users.

2.3.3 Business Rule Documentation

Business rules define the constraints, policies, and conditions that govern system behavior and user interactions within SkillLink. These rules ensure consistency, security, and fairness across the platform. To maintain a trusted and equitable marketplace, specific focus is placed on preventing bias and ensuring data integrity.

ID	Rule	Description	Enforcement Mechanism
BR-01	User Registration Rule	All users must register with a valid email address or phone number; accounts must be associated with exactly one primary role.	System-level validation during account creation; role-based access control (RBAC).
BR-02	Profile Completion Rule	Service Providers and Job Seekers must complete profiles, including skills and credentials, before eligibility for matching.	System prevents access to "Search" and "Apply" features until profile completion status is 100%.
BR-03	Verification Rule	Only verified Service Providers and Job Seekers can accept bookings or apply for jobs.	Administrative approval flag required in the database to unlock transaction-based use cases.
BR-04	Job & Service Posting Rule	Only Employers can post jobs, and only Customers can post service requests; mandatory fields include category, location, and budget.	Input validation forms with mandatory fields; role-restricted "Post" buttons.
BR-05	AI Matching & Anti-Bias Rule	Matches are generated based on skills, location, and performance while strictly	Attribute Masking and Neutral Data Weighting within the AI engine logic

		masking gender, age, or ethnicity.	to prevent algorithmic bias.
BR-06	Communication Rule	In-app messaging is enabled only after a post is active; video calls require mutual consent.	Communication channel is initialized only upon successful matching or query acceptance.
BR-07	Payment Rule	Payments must be processed through the platform via sandbox-based local gateways before confirmation.	Escrow-style fund holding; transaction status must reach "Confirmed" before service start.
BR-08	Rating and Review Rule	Ratings/reviews can only be submitted after a job or service is marked as completed.	Review form remains locked until the system detects a "Completed" status in the transaction log.
BR-09	Dispute Rule	Any transaction dispute must be handled by the Admin based on system logs and records.	"Dispute Resolution" workflow accessible via Admin Dashboard upon user flag.
BR-10	Admin Control Rule	Admins have full access to user management, verification, and reporting functions.	Secure Admin login with elevated privileges and audit logging for all actions.

Table 6: Business Rule Table

2.3.4 User Interfaces Prototype

2.3.4.1 Home Screen (Landing Page)

- **Purpose:** Serves as the central hub for all user types (Customer, Provider, Employer) to navigate the platform's core offerings.
- **Key Features:**
 - **Search Bar:** Allows users to immediately search for specific services or talents.
 - **Location Indicator:** Displays the user's current city (e.g., Addis Ababa) to facilitate location-based matching.
 - **Popular Services Grid:** Provides quick-access categories such as Plumbing, IT Support, Accounting, and Electrician.
 - **Primary Action Buttons:** Two large, color-coded buttons—"Find Workers Near You" (for customers) and "Find Jobs" (for seekers)—to direct users to their respective workflows.

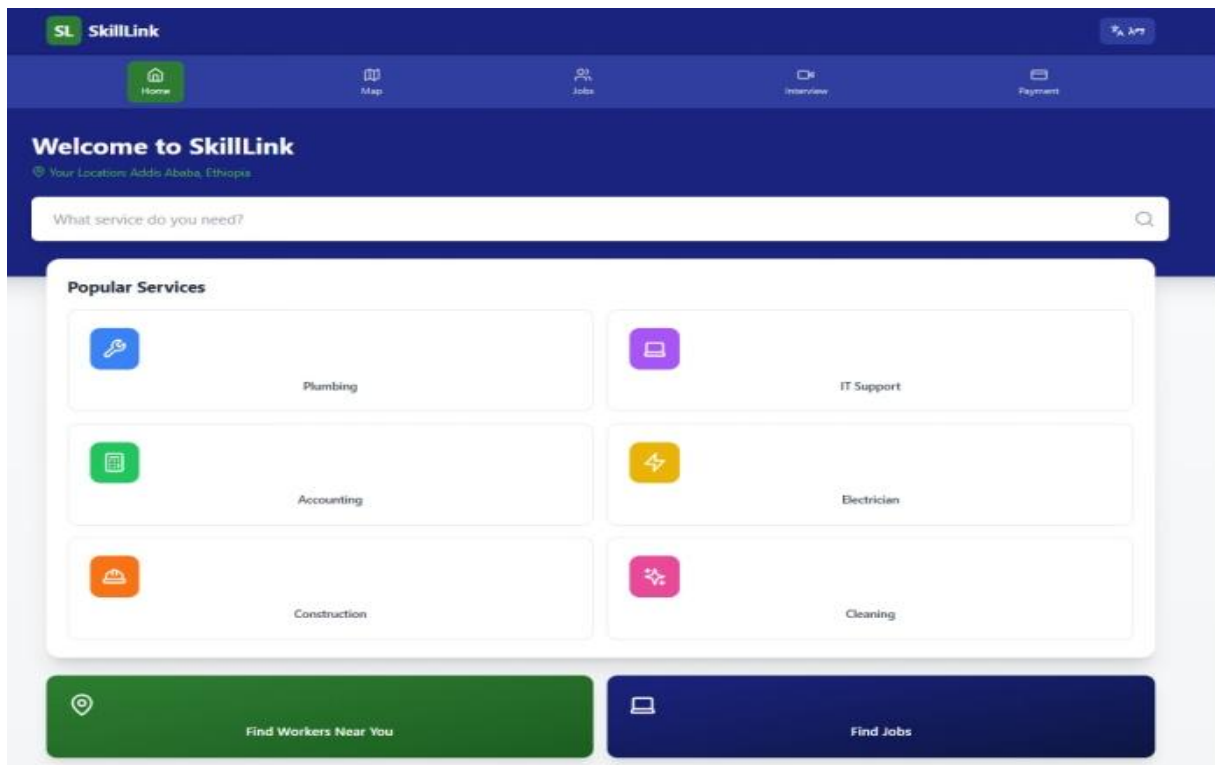


Figure 2: Home Page

2.3.4.2 AI-Matched Candidates (Jobs) Screen

- **Purpose:** Implements the "Intelligent Matching" objective by presenting a ranked list of verified talent tailored to specific job requirements.
- **Key Features:**
 - **Smart Ranking:** Displays candidates with a percentage-based "match score" (e.g., 96%, 92%) determined by the AI engine based on skills and performance data.
 - **Talent Cards:** Each card provides a snapshot of the professional, including their title, years of experience, educational background (e.g., AAU, Unity University), and specific skill tags (e.g., React, Financial Modeling).
 - **Direct Interview Link:** A prominent "Start Video Interview" button on each card facilitates the in-app video communication requirement (**FREQ-10**).
 - **Verification Visibility:** Integrates candidate profiles with their verified credentials and portfolios.

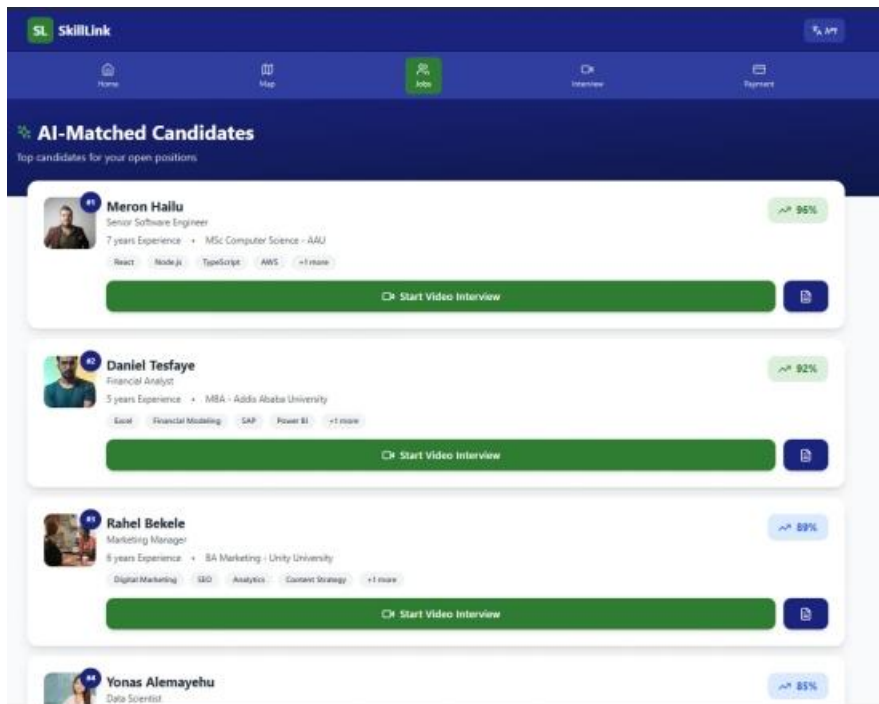
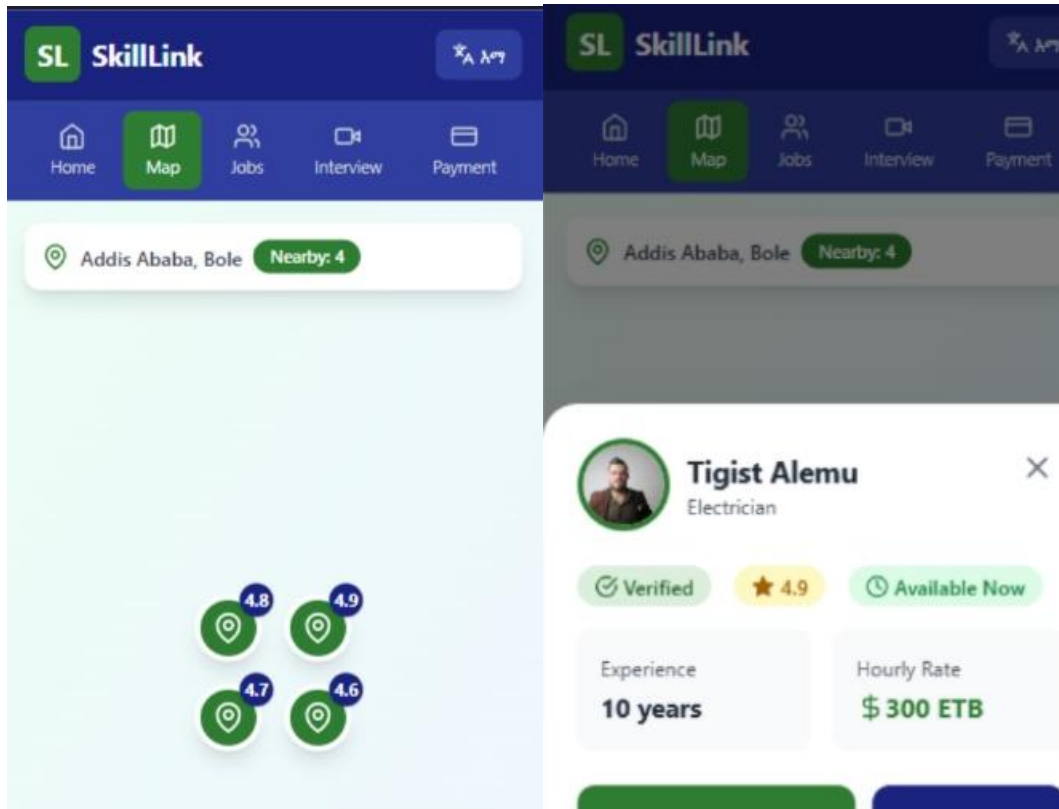


Figure 3: Job Page

2.3.4.3 Map Interface (Navigation Tab)

- **Purpose:** To provide a visual, geographic representation of available service providers or job opportunities.
- **Key Features:** Uses real-time geographic data to allow customers to visualize how close a service provider is to their location, reducing response times for on-demand services.



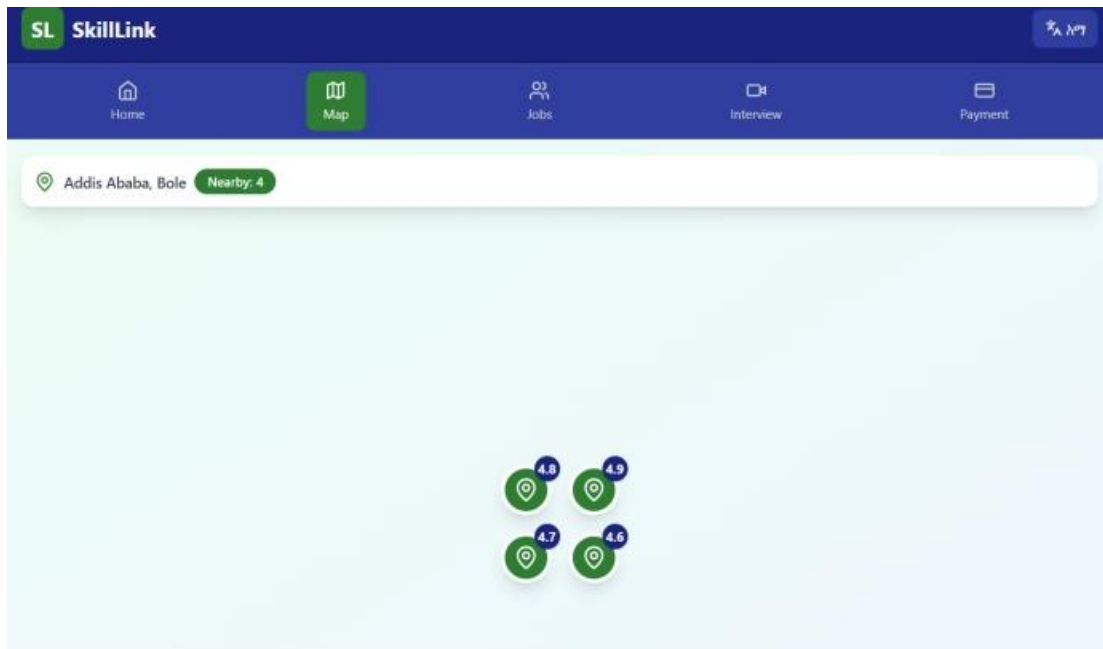


Figure 4: Map Interface (Navigation Tab)

2.3.4.4 Interview & Communication Module

- **Purpose:** Reduces friction in the hiring process by centralizing all pre-employment interactions.
- **Key Features:** Supports real-time messaging and video conferencing, allowing employers and job seekers to connect without leaving the platform.

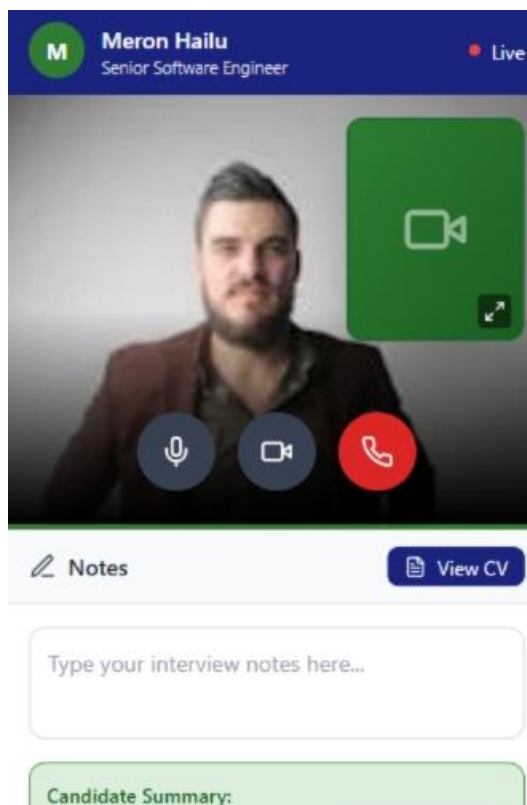
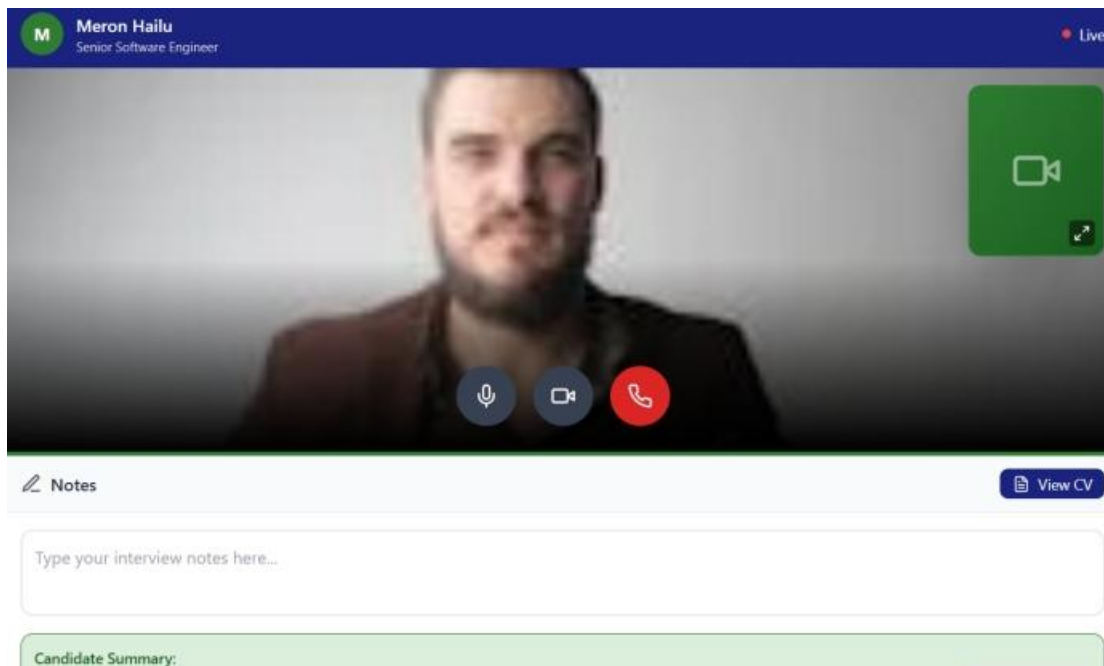


Figure 5: Interview & Communication Module

2.3.4.5 Payment & Transaction Dashboard

- **Purpose:** Ensures secure, transparent financial exchanges using integrated digital payment methods.
- **Key Features:** Tracks transaction history, manages service bookings, and facilitates payments through localized systems like mobile money.

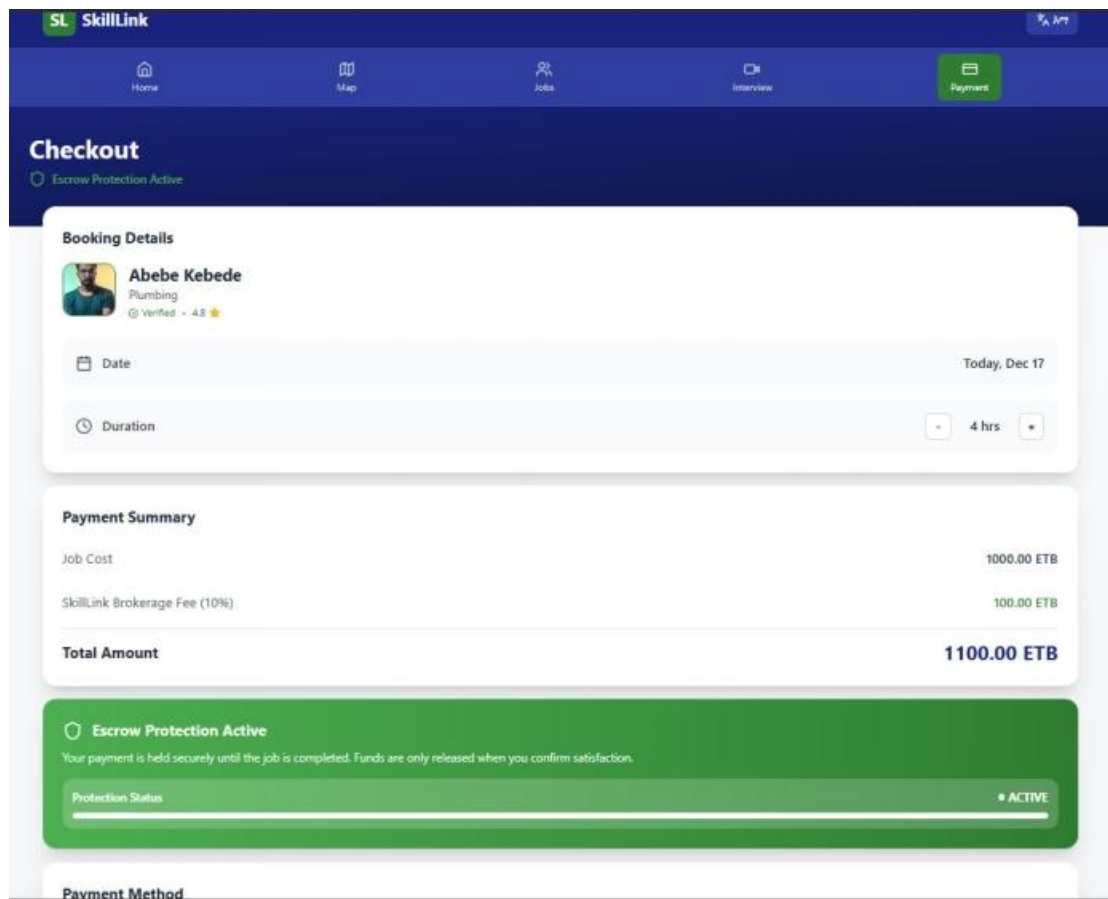


Figure 6: Payment Page

2.3.5 State chart diagram

The state chart diagram illustrates the dynamic behavior of the SkillLink system by showing how a job or service request transitions through different states over time.

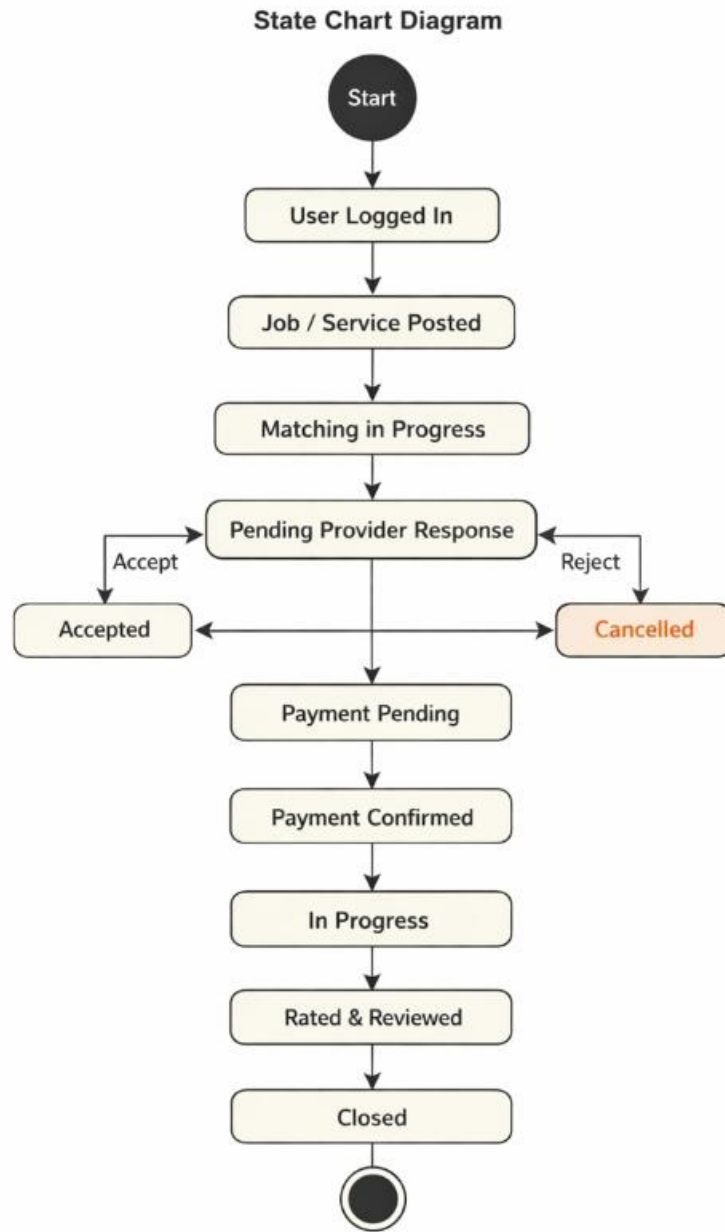


Figure 7: State Chart Diagram

2.3.6 Activity Diagram

Activity diagrams illustrate the workflow of operations performed by the system for each use case.

Activity Diagram Description (Job Recruitment / Service Booking)

1. User logs into the system.
2. User completes or updates profile.
3. Employer posts job or customer requests service.
4. System performs matching.
5. User reviews recommended matches.
6. Communication is initiated.
7. Selection is confirmed.
8. Payment is processed.
9. Job or service is completed.
10. Rating and review are submitted.
11. System updates records.

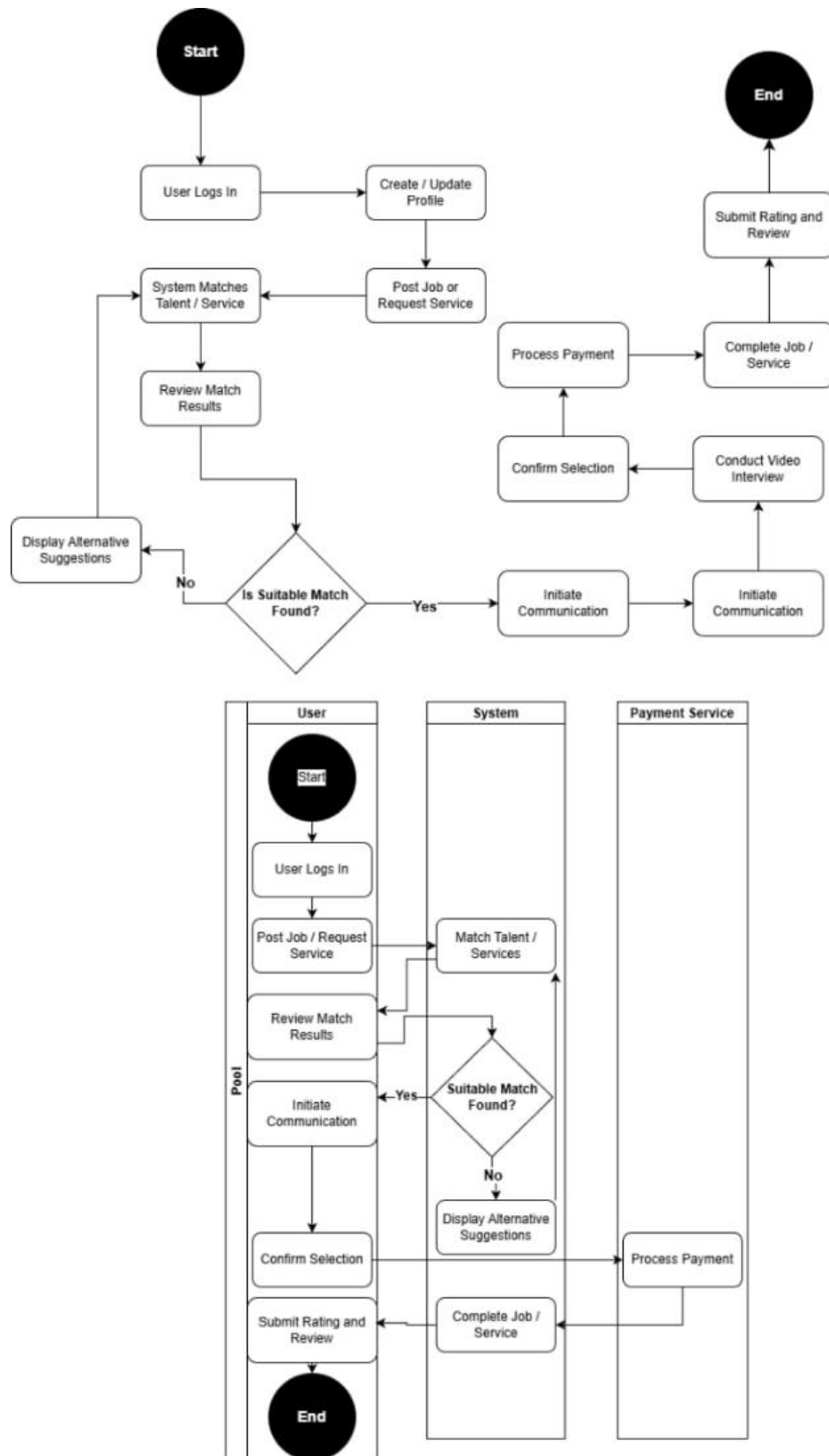


Figure 8: presents the activity diagram describing the workflow of job recruitment and taskflow

2.3.7 Sequence Diagram

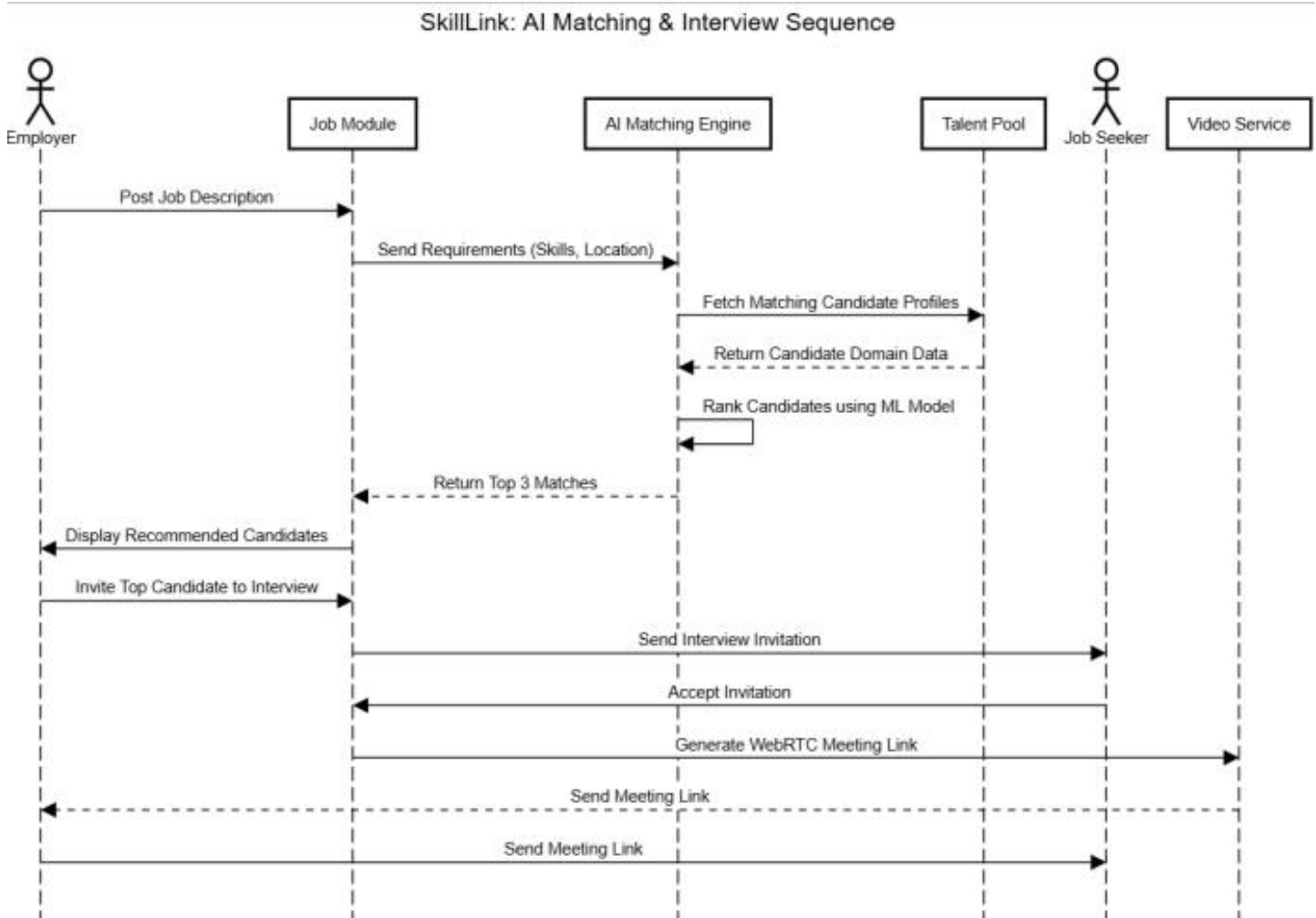


Figure 9: AI Matching & Interview Sequence

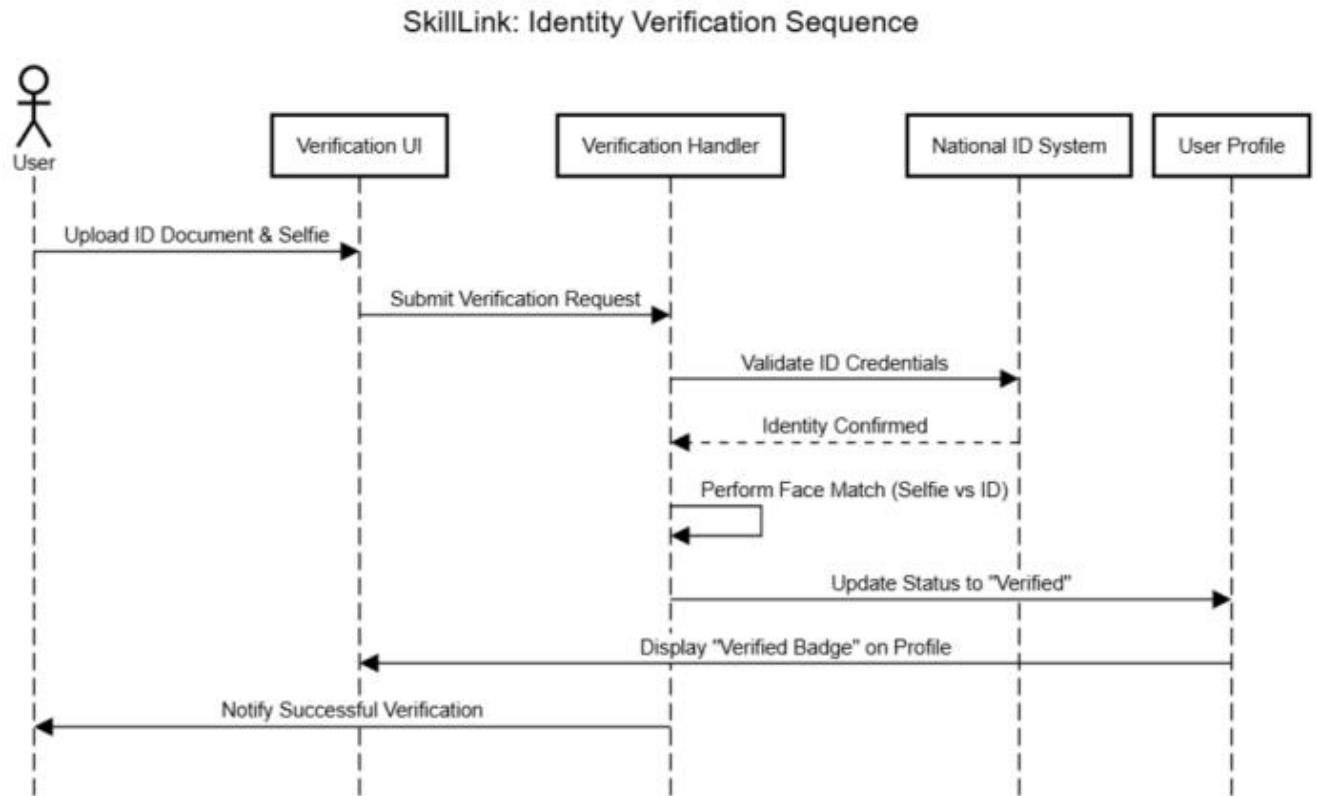


Figure 10: Identity Verification Sequence

SkillLink: Service Booking & Payment Sequence

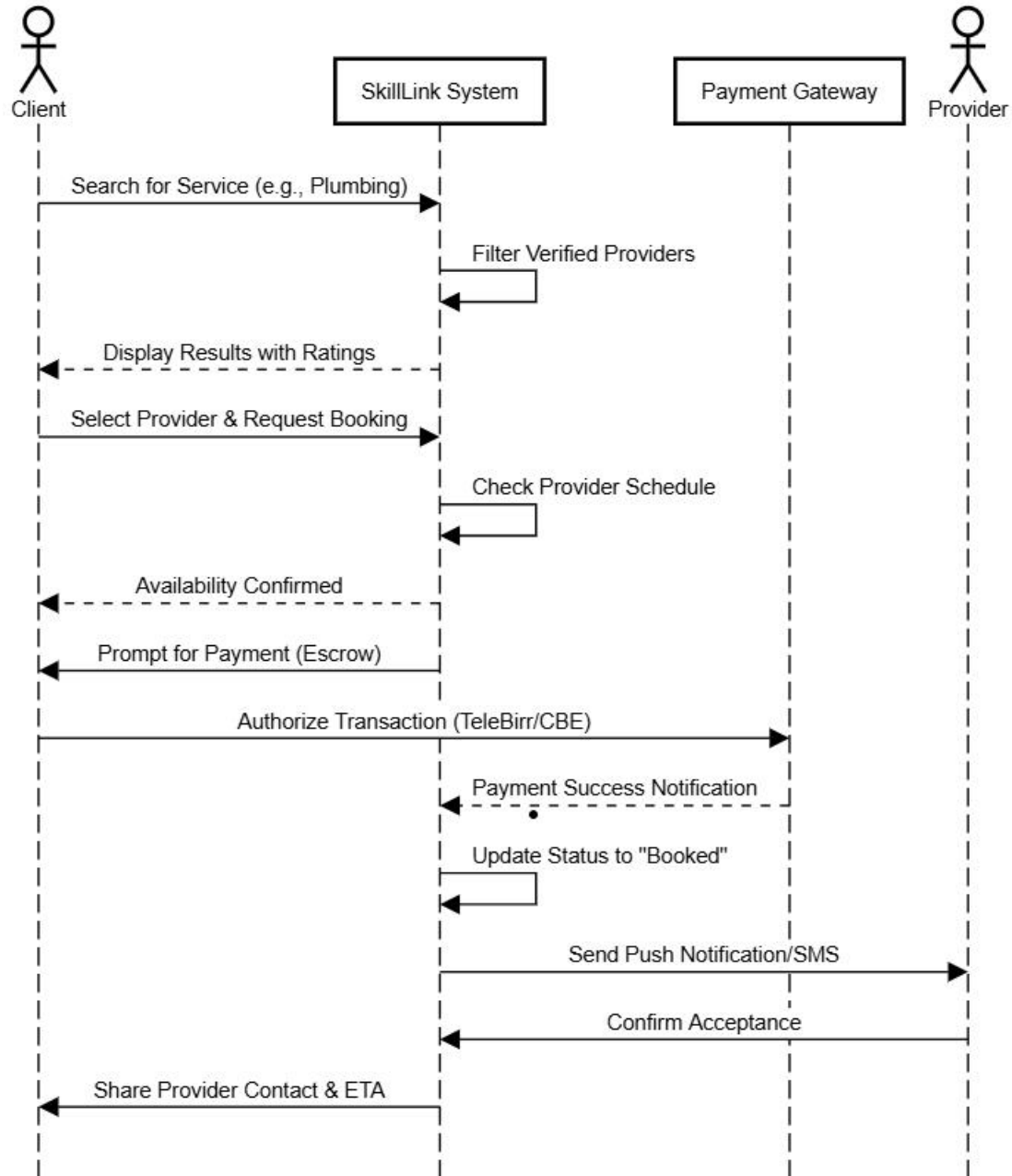


Figure 11: Service Booking & Payment Sequence

Figure 12: Analysis Class Model



2.3.9 Logic Model: SkillLink

1. Customer Logic Model

Objectives:

1. Browse services and view AI-recommended providers.
2. Book services and manage appointments.
3. Make secure payments via the platform.
4. Provide ratings and feedback for services rendered.

Workflow:

1. **Login & Dashboard:** Customer logs in; successful authentication directs them to a personalized dashboard featuring "Quick Book" options and AI-suggested providers based on location.
2. **AI-Powered Search:** - Customer enters a service need (e.g., "Plumbing").
 - ✓ The **AI Module** filters providers by skill, proximity, and rating.
 - ✓ The system displays a ranked list of the best matches.
3. **Booking & Escrow:** - Customer selects a Provider and chooses a time slot.
 - ✓ The system calculates the cost and prompts for payment.
 - ✓ Upon payment, the **Transaction** class holds funds in escrow until the job is confirmed complete.
4. **Feedback:** After the service status is marked "Complete," the Customer submits a 1-5 star rating and comment, which updates the Provider's `AvgRating`.

Pseudo-code

FUNCTION process_service_booking(customer_id, provider_id, service_details):

1. AI Recommendation Check

recommended_providers = AI_Engine.match(service_details.location, service_details.skill)

IF provider_id IN recommended_providers:

2. Payment & Escrow

total_cost = calculate_cost(service_details)

payment_status = Transaction.process_payment(customer_id, total_cost)

IF payment_status == "SUCCESS":

Transaction.hold_in_escrow(total_cost)

Notification.send(provider_id, "New Booking Request")

3. Provider Response

provider_action = Provider.respond_to_booking()

IF provider_action == "ACCEPT":

Booking.status = "CONFIRMED"

WAIT_FOR_COMPLETION()

ELSE:

Transaction.refund(customer_id)

Booking.status = "DECLINED"

ELSE:

PRINT"Payment Failed"

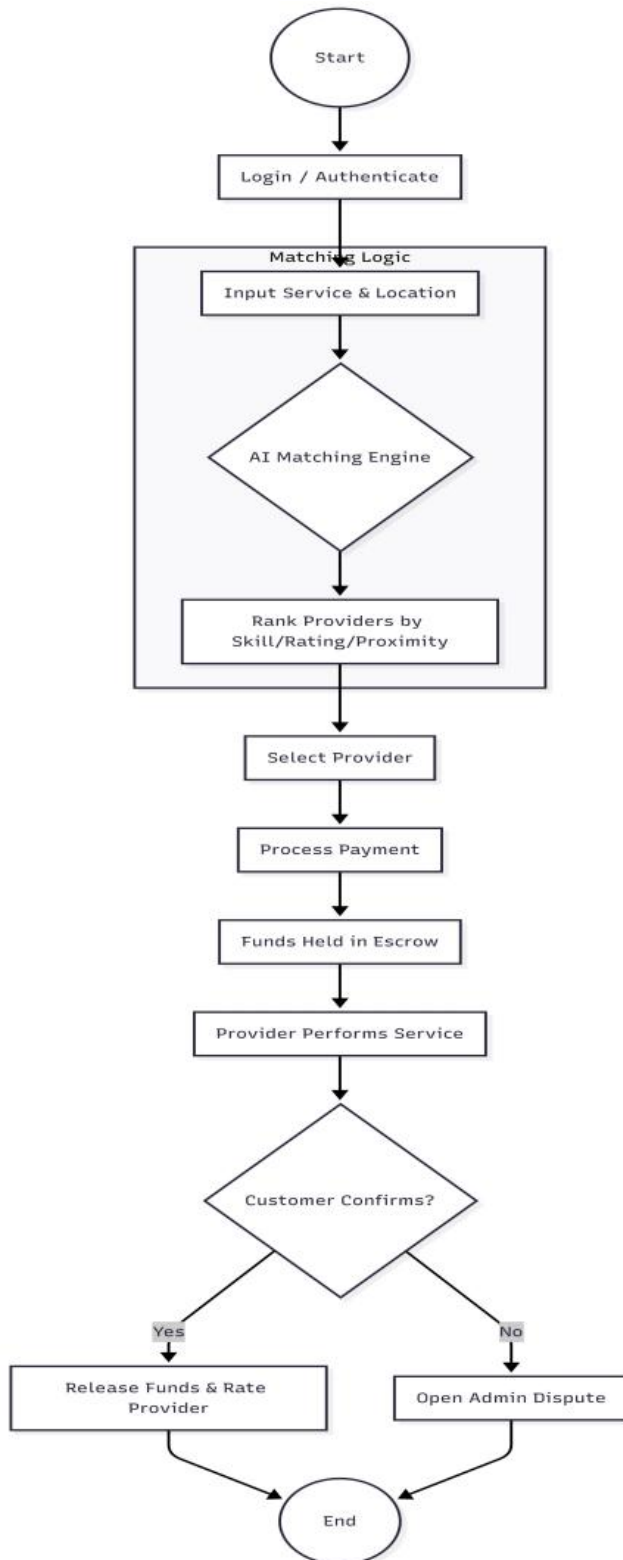


Figure 13: Customer Logic Model FlowChart

2. Provider Logic Model

Objectives:

1. Manage professional profile, skills, and portfolio.
2. Set availability and accept/reject service bookings.
3. Apply for long-term jobs posted by Employers.
4. Track earnings and requested payouts.

Workflow:

1. **Profile Management:** Provider updates their `SkillSet` and uploads portfolio images. The system tags these skills for the AI matching engine.
2. **Handling Bookings:** - When a Customer places a booking, the Provider receives a notification.
 - ✓ The Provider checks their schedule and chooses to **Accept** or **Decline**.
 - ✓ If Accepted, the system shares the job location and contact details.
3. **Job Application:** - Provider browses "Job Postings" created by Employers.
 - ✓ Provider clicks "Apply"; the system sends their verified profile and rating history to the Employer.
4. **Earnings:** Once a job is marked "Done" by the customer, the system triggers the **Transaction** logic to release funds from escrow to the Provider's digital wallet.

Pseudo-code

FUNCTION `employer_hiring_workflow(employer_id, job_requirements)`:

1. Create Posting

```
new_job = JobPosting.create(employer_id, job_requirements).
```

2. AI Scanning Logic

```
candidate_pool = Database.get_all_providers(verified=True)
```

```
ranked_candidates = []
```

```
FOR provider IN candidate_pool:
```

```
    match_score = AI_Engine.calculate_compatibility(job_requirements, provider.profile)
```

```
    ranked_candidates.add({provider.id, match_score})
```

3. Present Shortlist

```
shortlist = ranked_candidates.sort_by_score().top(10)
```

```
RETURN shortlist
```

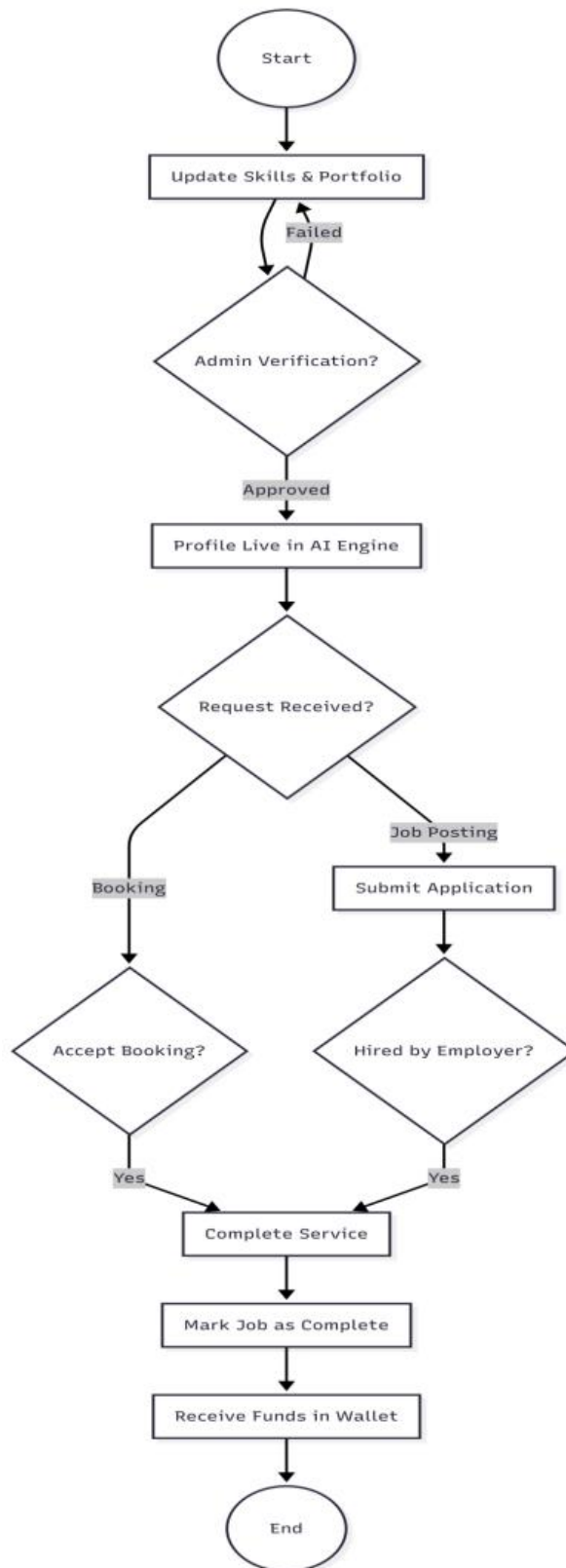


Figure 14: Provider Logic Model

3. Employer Logic Model

Objectives:

1. Post detailed job requirements and salary ranges.
2. Use AI to scan the database for qualified candidates.
3. Manage the hiring process (Interviewing/Hiring).

Workflow:

1. **Job Posting:** - Employer fills out the **JobPosting** form (Title, Required Skills, Budget).
 - ✓ The system saves the post and makes it visible to all qualified Providers.
2. **AI Candidate Matching:** - Employer selects "Find Matches."
 - ✓ The **AI Module** compares the `JobPosting` requirements against all `Provider` profiles.
 - ✓ The system presents a shortlist of the top 10 most compatible candidates.
3. **Hiring:** Employer reviews applications, conducts interviews, and selects a candidate. The system then initiates the contract/payment terms.

4. Admin Logic Model

Objectives:

1. Verify Provider credentials to ensure platform trust.
2. Monitor transactions and resolve disputes.
3. Manage platform-wide categories and analytics.

Workflow:

1. **Verification:** Admin reviews uploaded identity documents from new Providers. If valid, the Admin sets `IsVerified` to `True`.

2. **Dispute Resolution:** If a Customer reports a booking issue, the Admin accesses the **Transaction** and **Booking** logs to decide whether to refund the Customer or pay the Provider.

Pseudo-code

```
FUNCTION admin_verification_process(provider_id):
```

```
    docs = Provider.get_uploaded_documents()
```

```
    IF Admin.manually_verify(docs) == "VALID":
```

```
        Provider.IsVerified = True
```

```
        Notification.send(provider_id, "Your profile is now live.")
```

```
    ELSE:
```

```
        Provider.IsVerified = False
```

```
        Notification.send(provider_id, "Verification failed. Re-upload documents.")
```

```
FUNCTION resolve_dispute(transaction_id, customer_report):
```

```
    logs = Database.get_logs(transaction_id)
```

```
    evidence = Booking.get_completion_proof()
```

```
    IF evidence == "VALID":
```

```
        Transaction.release_escrow_to_provider()
```

```
    ELSE:
```

```
        Transaction.refund_to_customer()
```

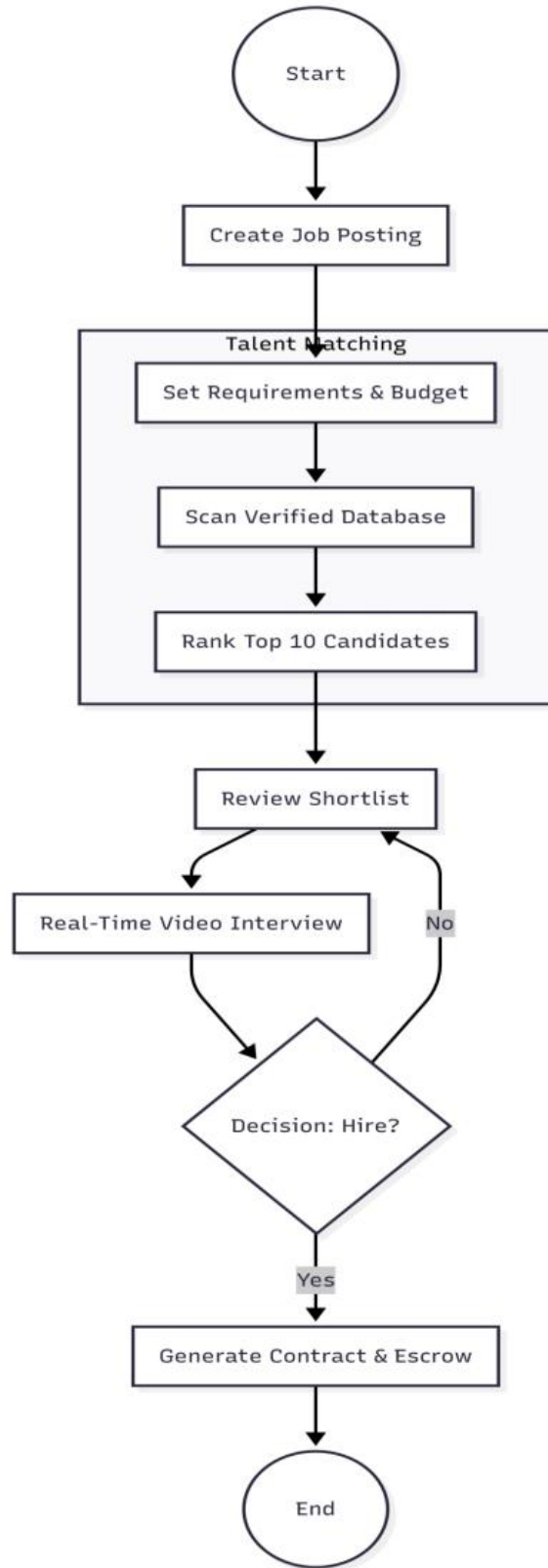



Figure 15: Admin Logic Model FlowChart

2.4 Non functional Requirement

Non-functional requirements describe the quality attributes and constraints of the SkillLink system to ensure optimal performance, security, usability, and reliability.

Non-Functional Requirements List	NFR ID	Requirement	Measurable Metric / Target
Performance	NFR-01	System shall respond to user requests and perform searches efficiently.	Response time ≤ 2 seconds under moderate load (up to 1000 concurrent users); search/matching ≤ 3 seconds.
Security	NFR-02	System shall protect user data and enforce secure access.	HTTPS with TLS 1.2+, AES-256 encryption for sensitive data, JWT authentication, lock account after 5 failed logins, password complexity: ≥ 8 characters with upper/lowercase, number, special character.
Usability	NFR-03	System shall be intuitive and easy to use for all user types.	Core tasks (registration, posting, applying) completed within 3–5 clicks; guidance, tooltips, and error messages provided.
Availability	NFR-04	System shall be accessible to users with minimal downtime.	Availability $\geq 99.5\%$; scheduled maintenance ≤ 2 hours/month with prior notice.
Scalability	NFR-05	System shall support growth in users and features without major redesign.	Support ≥ 5000 concurrent users; modular architecture allows addition of new roles/services with minimal impact.
Reliability	NFR-06	System shall maintain data integrity and	100% transaction logging; daily backups; automatic recovery within 30 minutes of failure.

		recover from failures.	
Compatibility	NFR-07	System shall work across different devices and browsers.	Fully functional on Chrome, Edge, Firefox, Safari (latest versions); supports mobile screen widths 320px–1440px.
Localization	NFR-08	System shall support multiple languages and payment methods.	Full UI in English and Amharic; support local payment systems (M-Birr, CBE Birr) and international cards.

Table 7: Non functional Requirement

2.5 System Requirement

2.5.1 Hardware Requirements

The hardware specifications are categorized based on whether the system is being actively developed or being deployed for end-users.

1. Development Requirements (Local Workstation)

These are necessary for the development team to build and test the AI-driven smart matching platform.

- **CPU:** Intel Core i5 (8th Gen) or AMD Ryzen 5 equivalent as the minimum requirement. For faster AI model training, an Intel i7 or Ryzen 7 is recommended.
- **RAM:** 8 GB DDR4 minimum; 16 GB is recommended for running Docker containers and development environments simultaneously.
- **Storage:** 256 GB SSD minimum for fast indexing; 512 GB NVMe SSD recommended for large datasets and portfolios.
- **GPU:** Dedicated NVIDIA GeForce (optional) to accelerate local AI matching model training.

2. Production/Client Requirements (End-User Access)

The hardware needed by users to access the SkillLink services.

- **Mobile/Desktop:** Any modern smartphone or PC capable of running a web browser.
- **Camera:** Standard integrated webcam or smartphone camera (Minimum 720p) for Identity Verification and Video Interviews.
- **Connectivity:** Stable internet connection (Minimum 2-5 Mbps) for real-time communication and localized payment processing.

2.5.2 Software Requirements

The software stack is divided into the tools used for building the system and the cloud infrastructure used for deployment.

1. Development Environment (Build & Test)

- **OS:** Windows 10/11, macOS Monterey, or Linux (Ubuntu 22.04 LTS).
- **IDE:** Visual Studio Code with ESLint for code quality.
- **Runtime:** Node.js (v18.x or higher) for the backend and JavaScript/TypeScript for frontend logic.
- **Database:** PostgreSQL (Structured transactions) or MongoDB (Flexible user profiles).
- **AI Tools:** Python 3.10+ with Scikit-learn for the skill-matching engine.
- **Testing:** Postman for API validation of payment and matching endpoints.

2. Production Environment (Cloud Hosting & Deployment)

To ensure high availability and scalability, the following cloud options are proposed:

- **Frontend Hosting:** Vercel is recommended for Next.js deployments due to its optimized global CDN.
- **Application Hosting:** AWS (Amazon Web Services) using EC2 or Lambda for scalable backend services.
- **Database Cloud:** MongoDB Atlas or AWS RDS for managed database services with automated backups.
- **Communication:** Daily.co or Twilio Video API for the integrated WebRTC video interview module.
- **Security:** JWT (JSON Web Tokens) for stateless authentication and SSL certificates for secure HTTPS data transmission.

Summary of Multiplicity and Roles in System Integration

As defined in the Conceptual Class Diagram, the software requirements facilitate the following associations:

- **Vercel/Next.js** handles the Customer role interaction (multiplicity 1 to many Bookings).
- **Stripe/Local Payment APIs** manage the Transaction domain object (multiplicity 1 to 1 with Bookings).
- **AWS AI Services** process the JobPosting and Service entities to generate Match scores.

2.5.3 System Architecture Diagram

This diagram illustrates how the hardware and software components interact to fulfill the project's objectives.

The diagram above represents the **SkillLink Architecture**, where:

1. **Frontend (UI):** Connects users (Customers, Providers, Employers) to the system.
2. **Logic Layer (Backend):** Processes requests and handles the business logic.
3. **AI Module:** Takes data from the database to generate smart recommendations.
4. **Integration Layer:** Connects to secure third-party payment gateways.

2.6 Key Abstraction with CRC Analysis

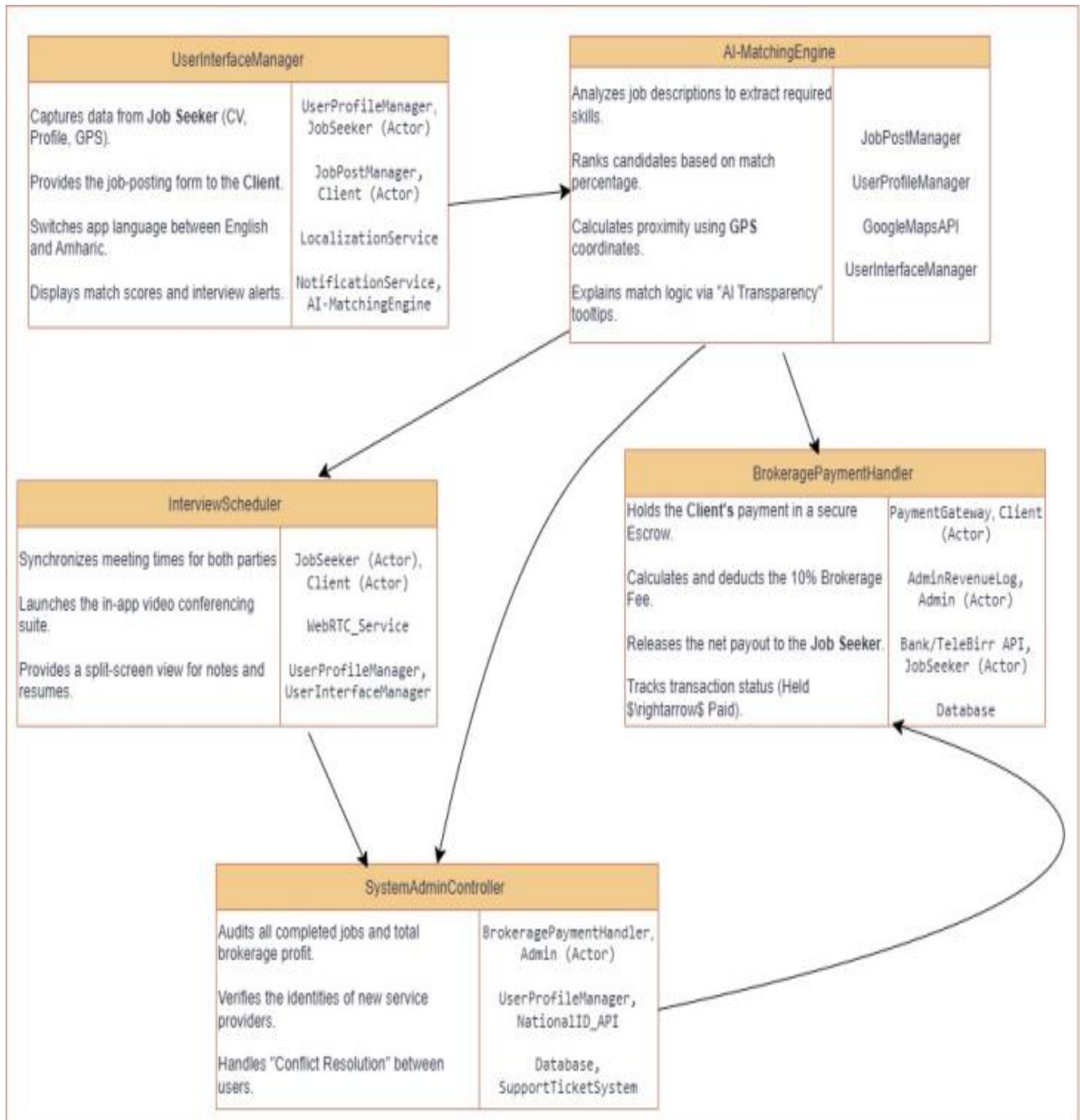


Figure 16: CRC Analysis Diagram

2.6.1 Conceptual Modeling: Class Diagram

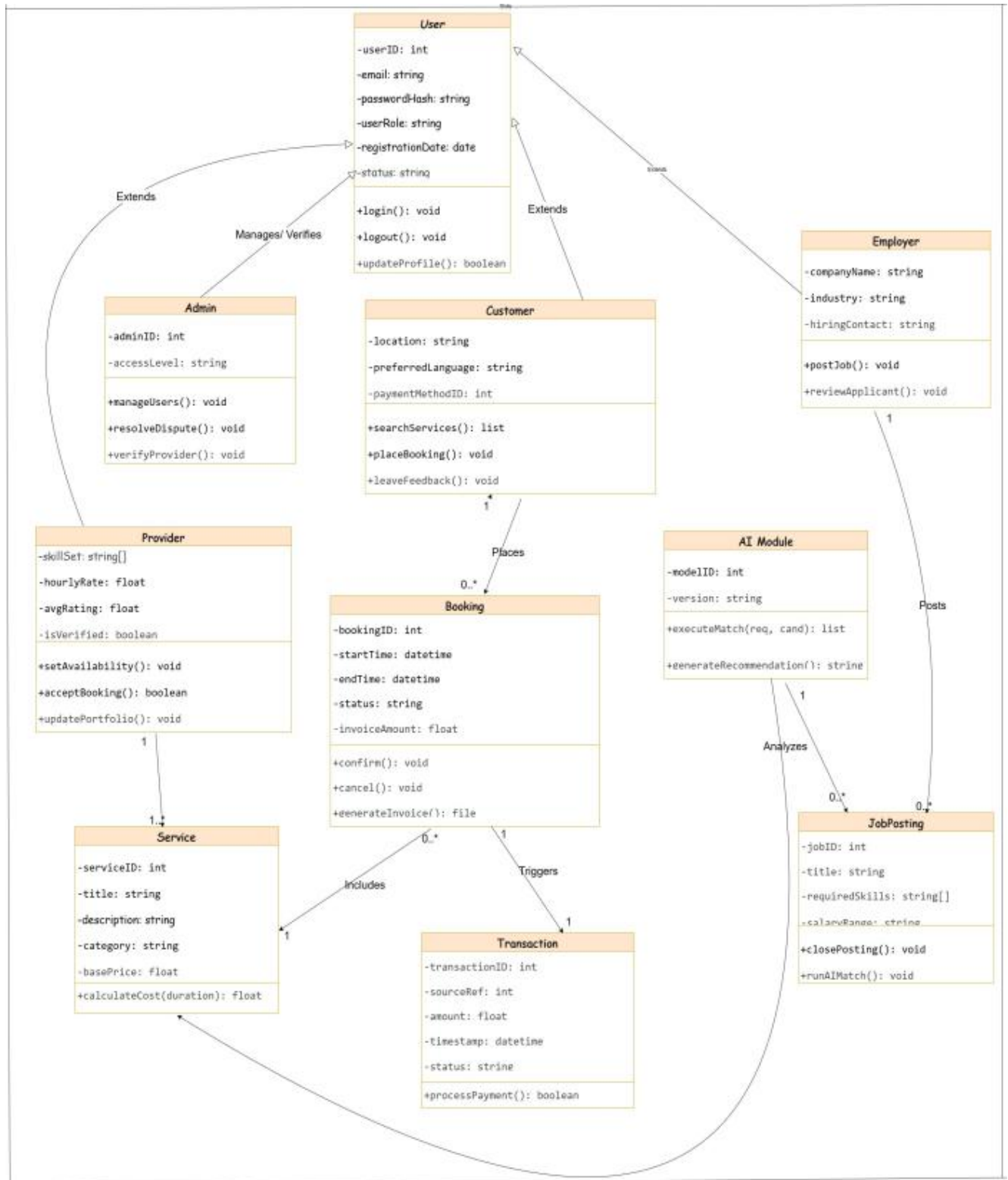


Figure 17: Class Diagram

2.6.2 Identifying Change Cases

Change cases evaluate the system's flexibility for future requirements without requiring a complete redesign.

Change Case	Description	Affected Class	Likelihood	Impact
New Payment Gateways	Adding CBE Birr or International Card support.	BrokeragePaymentHandler	High	High
National ID Integration	Automating verification via Ethiopia's Fayda API.	UserProfileManager	Medium	High
USSD Support	Enabling offline job applications via shortcodes.	ManagerInterface	Medium	Medium
Expanded Localization	Adding regional languages (e.g., Oromiffa, Tigrinya).	LocalizationService	High	Low
AI Evolution	Upgrading the matching algorithm for better accuracy.	AI-MatchingEngine	High	Medium

Table 8: Identifying Change Cases

2.6.3 User Interface Prototyping (The Process)

The prototyping process for **SkillLink** followed an iterative **User-Centered Design (UCD)** methodology. This ensured that the platform remains accessible to both high-level corporate recruiters and local daily laborers.

Step 1: Requirements & User Persona Mapping

We began by defining two distinct "User Journeys" to dictate the interface logic:

- **The "Emergency" Path:** Designed for a client needing a plumber immediately. The process focused on **GPS-first navigation** and Amharic labeling to reduce cognitive load during a crisis.

- **The "Career" Path:** Designed for long-term recruitment. This required a **data-centric dashboard** focused on AI Match Scores and professional CV visualization.

Step 2: Information Architecture (IA) & Wireframing

Before applying colors, we created the skeletal structure. We mapped out the "3-Click Rule"—ensuring that any user could find a service or post a job in three clicks or less.

- **Key Decision:** We placed the **Language Toggle** in the top-right header as a persistent element, allowing users to switch between English and Amharic at any stage of a transaction without losing data.

Step 3: Visual Identity & Design System

We developed a consistent **Design System** to ensure the app feels like a single, unified brokerage platform:

- **Typography:** We paired **Inter** (for English) with **Nyala/Abyssinica SIL** (for Ethiopic) to ensure that Amharic text is just as legible as English text on small mobile screens.
- **Color Psychology:**
 - * **Navy Blue (#1A237E):** Used for headers to establish corporate trust and stability.
 - **Emerald Green (#2E7D32):** Used for "Book Now" and "Pay" buttons to encourage action.
 - **Amber (#FFA000):** Used for "Escrow/Pending" status to signal caution and security.

Step 4: High-Fidelity Iteration (The Final Mockups)

The final stage involved converting wireframes into the high-fidelity images seen in the next section. During this phase, we integrated:

- **Real-time Feedback Loops:** Adding progress steppers (e.g., *e.g., Payment—->Escrow—->Release*) so users understand the brokerage logic visually.

- **GPS Overlay:** Integrating a custom-styled map interface that highlights "Verified" providers with blue badges.

Step 5: Prototyping Tools Used

- **Figma:** Used for creating the interactive layout and component library.
- **Adobe Color:** Used to ensure the color palette met **WCAG Accessibility Standards** for users with visual impairments.
- **Diagrams.net:** Used for mapping the user flow logic before the UI was rendered.

Chapter 3: System Design

This chapter presents the purpose and design goals of the **SkillLink platform – A Smart Platform for Connecting Local Talent and On-Demand Services**. The system is designed to address the challenges of connecting skilled individuals with customers, employers, and service seekers through a centralized digital platform. The design follows modern software engineering principles to ensure scalability, security, reliability, and usability.

Purpose of the System

The purpose of the SkillLink system is to provide a centralized and intelligent platform that enables efficient interaction between service providers, job seekers, employers, customers, and administrators. By integrating AI-based matching and secure transaction management, the system aims to:

1. **Facilitate Efficient Talent and Service Matching:** Enable customers and employers to quickly find suitable service providers or job seekers based on skills, availability, location, and ratings.
2. **Enhance Employment and Service Accessibility:** Provide a digital marketplace where local talent can access job and service opportunities easily through web and mobile platforms.
3. **Improve Decision Making:** Use AI-driven recommendations and ranking mechanisms to assist users in selecting the most appropriate candidates or services.
4. **Ensure Secure and Reliable Transactions:** Provide secure authentication, role-based access control, and safe payment processing to protect user data and financial transactions.

Design Goals

1. Performance

- **Scalability:** The system is designed to handle a growing number of users, job postings, and service requests without degradation in performance.

- **Speed:** Requests such as job searches, service matching, and profile retrieval are optimized for fast response times.
- **Efficiency:** Backend services and AI modules are designed to process requests efficiently, even during peak usage periods.

2. Dependability

- **Reliability:** Robust error-handling and logging mechanisms ensure consistent system operation.
- **Security:** Sensitive user data and transactions are protected using encryption and role-based access control (RBAC).
- **Availability:** The system ensures continuous availability through proper server management and fault-tolerant design.

3. End-User Experience

- **Usability:** The system provides intuitive and easy-to-use interfaces for all user roles.
- **Accessibility:** The platform supports both web and mobile access to ensure wide usability.
- **Personalization:** Users receive tailored job and service recommendations based on their profiles and activity.

4. Maintainability

- **Modularity:** The system is designed using modular components, making it easier to maintain and extend.
- **Documentation:** Comprehensive documentation supports system maintenance and future enhancements.
- **Testability:** Unit and integration testing are supported to ensure system correctness.

5. Compliance and Ethics

- **Data Privacy:** User data is handled in compliance with data protection and privacy standards.
- **Ethical AI:** The AI matching module is designed to reduce bias and ensure fairness and transparency.

Key Features Supporting Design Goals

1. **Centralized Database:** A unified data store for user profiles, job listings, service requests, and transactions.
2. **AI-Based Matching Module:** Intelligent recommendation of suitable service providers and job seekers.
3. **Multi-Platform Support:** Web and mobile applications for broad accessibility.
4. **Secure Data Management:** Authentication, authorization, encryption, and logging mechanisms ensure data security.
5. **Integrated Payment System:** Secure handling of payments and transaction records.

3.1 Architectural Design

System Architecture Overview

The SkillLink platform adopts a **layered client–server architectural design** to ensure modularity, scalability, and maintainability. The system architecture is composed of the following major components:

- **Presentation Layer**
- **Backend / API Layer**
- **Database Layer**
- **AI Matching Module**
- **Security and Logging Subsystem**
- **External Services Integration**

The architecture follows a service-oriented approach where the frontend communicates with backend services through secure RESTful APIs.

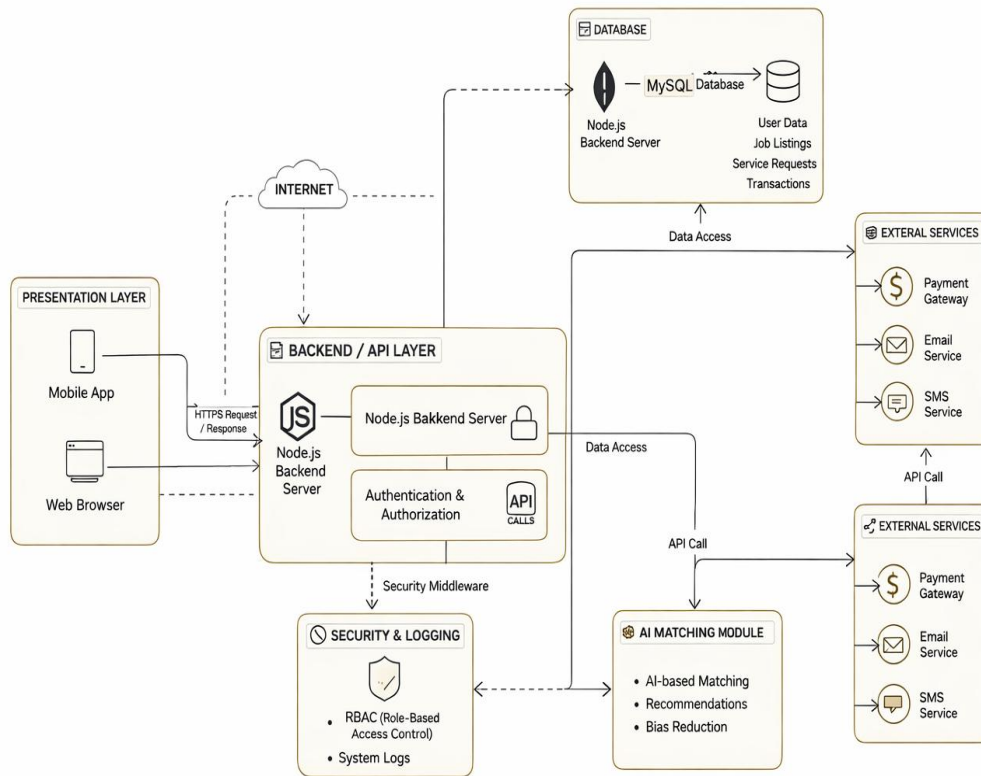


Figure 18: Architectural Design of the SkillLink System

Explanation of Component Interactions

1. Presentation Layer (Web & Mobile Applications)

- ✓ The presentation layer represents the user-facing part of the system and includes a web application and a mobile application.
- ✓ It provides interfaces for customers, service providers, job seekers, employers, and administrators.
- ✓ The frontend communicates with the backend server using secure HTTPS requests and receives responses in JSON format.
- ✓ User authentication requests, job searches, service requests, and profile management operations are initiated from this layer.

2. Backend / API Layer (Node.js Server)

- ✓ The backend server acts as the core processing unit of the system.
- ✓ It handles business logic, request validation, and coordination between system components.
 - Key responsibilities include:
 - User authentication and authorization
 - Job and service request management
 - AI matching request handling
 - Payment processing coordination
 - Communication with external services
- ✓ The backend enforces security policies and ensures role-based access control.

3. Database Layer

- ✓ The database layer stores all persistent system data, including:
 - User profiles and credentials
 - Job listings and service requests
 - Transaction records
 - Ratings and reviews
 - System logs
- ✓ The database interacts exclusively with the backend server to perform CRUD (Create, Read, Update, Delete) operations.
- ✓ Data integrity and consistency are maintained through structured schemas and constraints.

4. AI Matching Module

- ✓ The AI matching module analyzes user data, job requirements, and service requests to generate intelligent recommendations.
- ✓ It uses data such as skills, location, availability, and historical performance to rank candidates and services.

- ✓ The module communicates with the backend server through internal API calls. Bias reduction techniques are applied to ensure fair and ethical recommendations.

5. Security and Logging Subsystem

- ✓ This subsystem handles system-wide security mechanisms.
- ✓ It implements authentication and authorization using role-based access control (RBAC).
- ✓ All sensitive data transmissions are encrypted.
- ✓ System activities and errors are recorded through logging for auditing and monitoring purposes.

6. External Services Integration

- ✓ The system integrates with external services such as:
 - Payment gateways for transaction processing
 - Email services for notifications
 - SMS services for alerts and confirmations
- ✓ These services are accessed through secure APIs managed by the backend server.

3.1.1 Component Modeling

Component modeling describes the decomposition of the SkillLink system into modular and reusable software components and defines how these components interact to deliver system functionality. This approach improves scalability, maintainability, and system clarity by separating concerns across functional layers.

SkillLink follows a layered service-oriented architecture consisting of frontend presentation components, backend business logic components, AI processing modules, database services, and external system integrations.

3.1.1.1 Major System Components

The SkillLink system is composed of the following components:

1. User Interface Component

This component manages all user interactions with the system. It displays system information and captures user input for actions such as registration, job posting, service booking, communication, and payment.

Responsibilities:

- Render web pages and forms
- Validate client-side inputs
- Send requests to backend services
- Display system feedback and notifications

2. Authentication & Authorization Component

This component ensures secure access to the platform by managing user authentication and enforcing role-based access control.

Responsibilities:

- User login and registration
- JWT token management
- Role verification and access control

3. User Management Component

This component manages user profiles, credentials, verification status, and user ratings.

Responsibilities:

- Profile creation and updates
- Document storage
- Verification tracking

- Rating and review management

4. Job & Service Management Component

This component handles job postings, service listings, bookings, scheduling, and workflow status transitions.

Responsibilities:

- Job and service creation
- Booking management
- Status tracking (Pending, Confirmed, Completed)
- History management

5. AI Matching Component

This component provides intelligent recommendations by analyzing user skills, location, availability, and performance history.

Responsibilities:

- Calculate match scores
- Rank candidates and providers
- Return optimized recommendations

6. Communication Component

This component enables real-time chat and video interviews between users.

Responsibilities:

- Text messaging
- Video call sessions
- Communication history storage

7. Payment & Transaction Component

This component manages secure financial operations and transaction workflows.

Responsibilities:

- Payment initiation and confirmation
- Escrow handling
- Refund processing
- Transaction history management

8. Database Component

This component stores all persistent system data.

Responsibilities:

- User records
- Job and service records
- Payment transactions
- Communication logs
- System audit logs

9. Admin & Monitoring Component

This component supports administrative oversight and system governance.

Responsibilities:

- User verification
- Dispute handling
- Reporting and analytics
- System auditing

3.1.1.2 Component Interaction Description

The interaction between components follows a client-server architecture:

1. Users interact with the system through the User Interface Component.
2. Requests are forwarded to the Backend API Gateway.
3. The Backend routes requests to the appropriate business components:
4. Authentication Component for login and security
5. User Management Component for profile operations
6. Job & Service Management Component for postings and bookings
7. AI Matching Component for recommendations
8. Payment Component for financial transactions
9. Communication Component for chat and interviews
10. All business components store and retrieve data through the Database Component.

External services such as payment gateways and video APIs are accessed through the Integration Layer.

This modular architecture improves scalability, security, fault isolation, and future extensibility.

3.1.1.3 Component Diagram

The SkillLink component diagram shows the system’s modular architecture, divided into user interface, business logic, specialized services, and data layers, illustrating clear dependencies and well-defined component interactions.

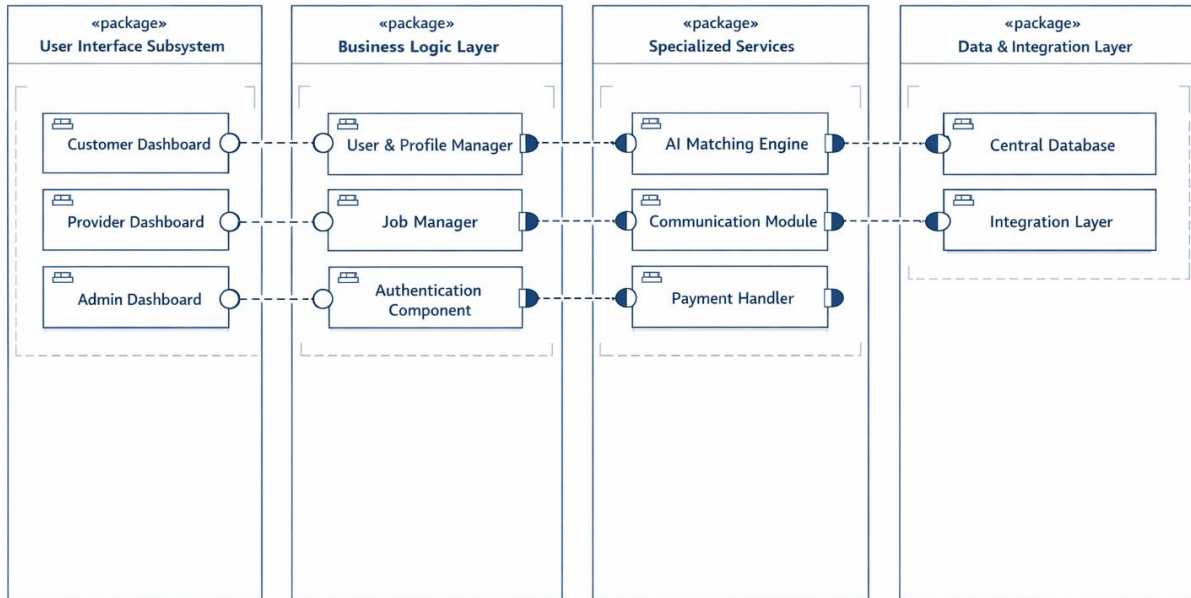


Figure 19: Component Diagram

3.1.2 Deployment Model

“The deployment diagram illustrates how the system components are distributed across physical nodes. Client applications run on user devices, while backend services are hosted on an application server. AI-related components are deployed on a dedicated AI server, and data storage is handled by a separate database server. Secure communication is maintained using HTTPS and authenticated service calls.”

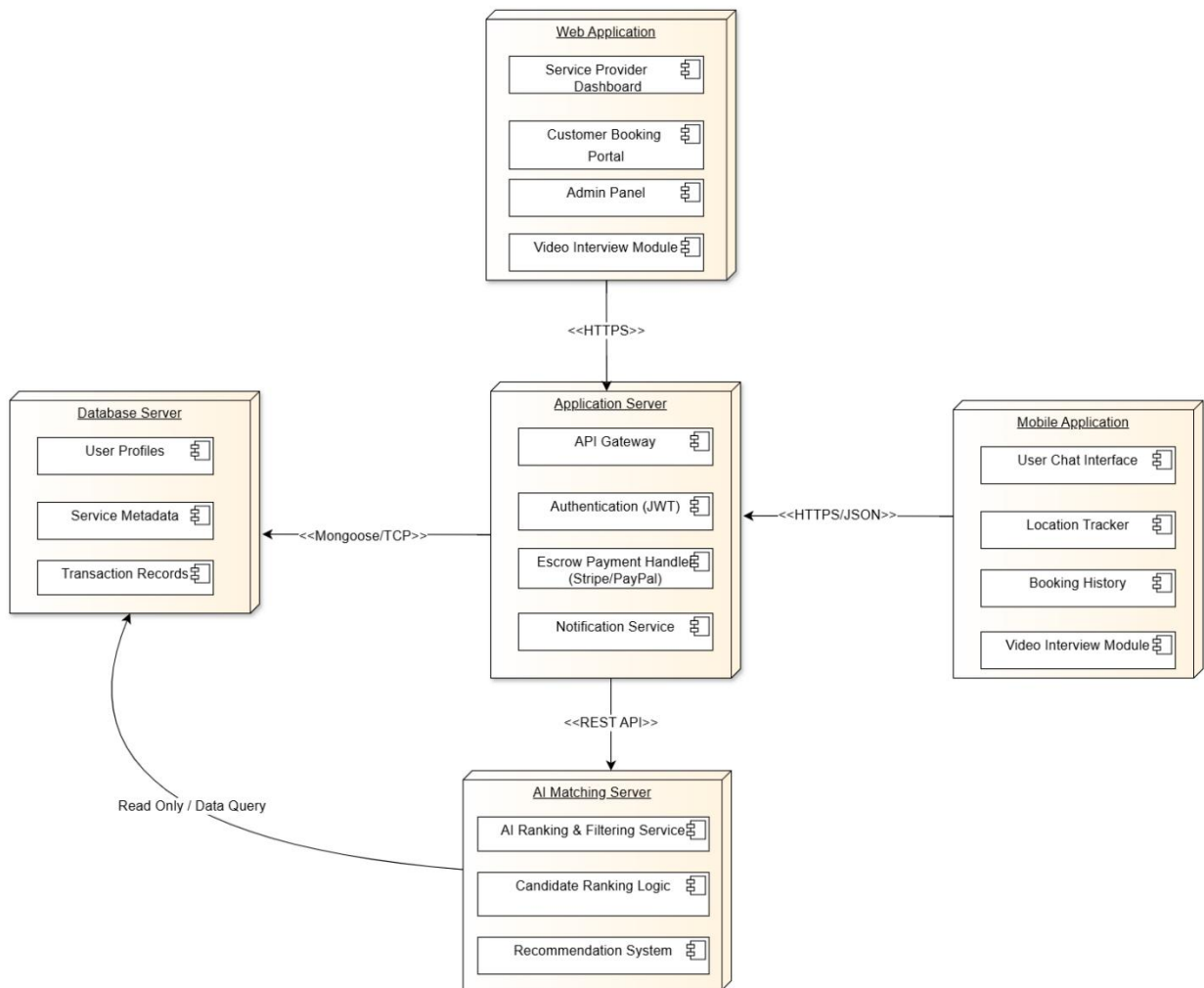


Figure 20: Deployment Diagram

3.2 – Detail Design

3.2.1 Design class model

Figure : Shows the object-oriented structure of the system, including classes, attributes, methods, and relationships (inheritance, associations) between User roles, Jobs, Services, and AI module.

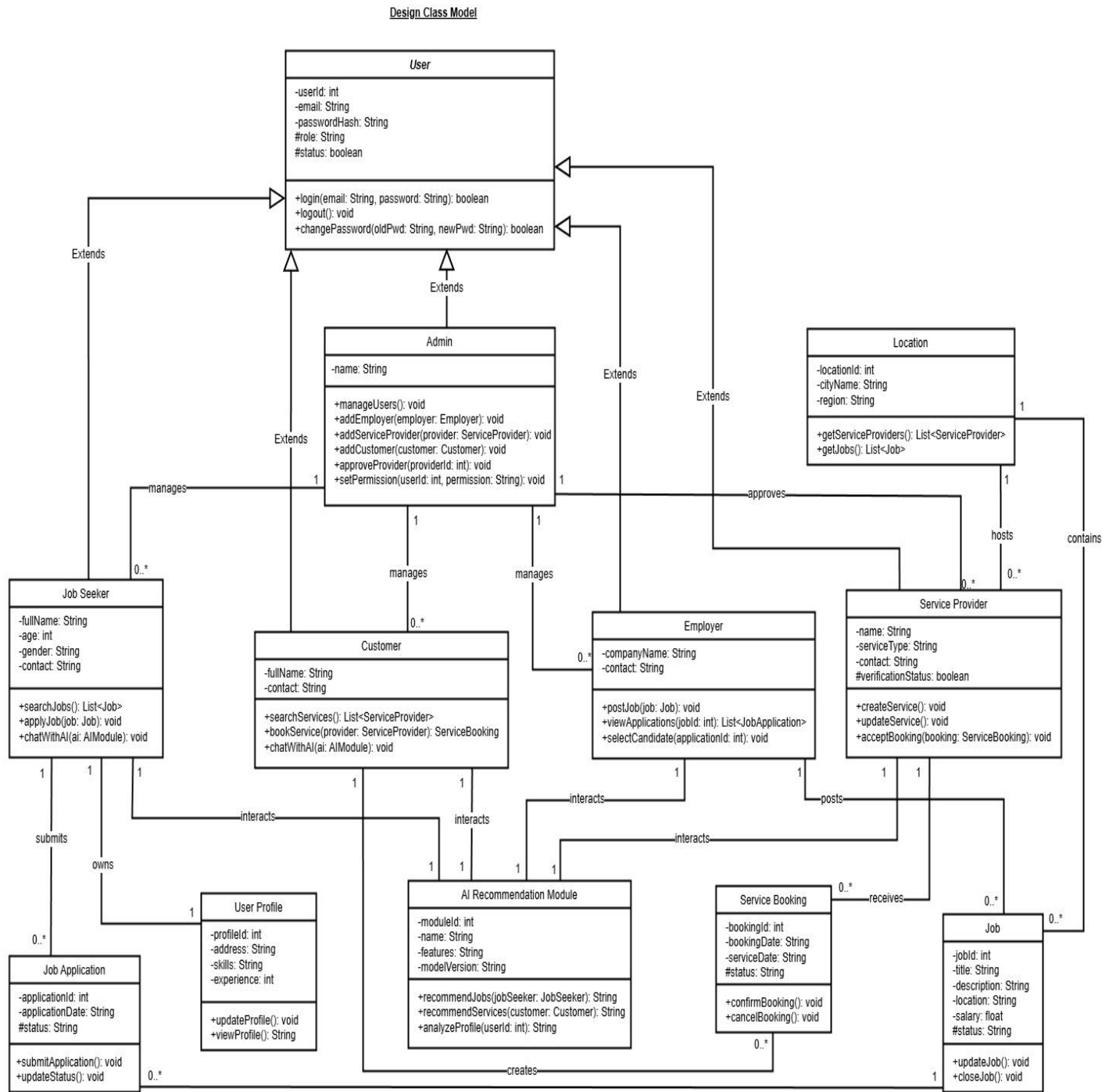
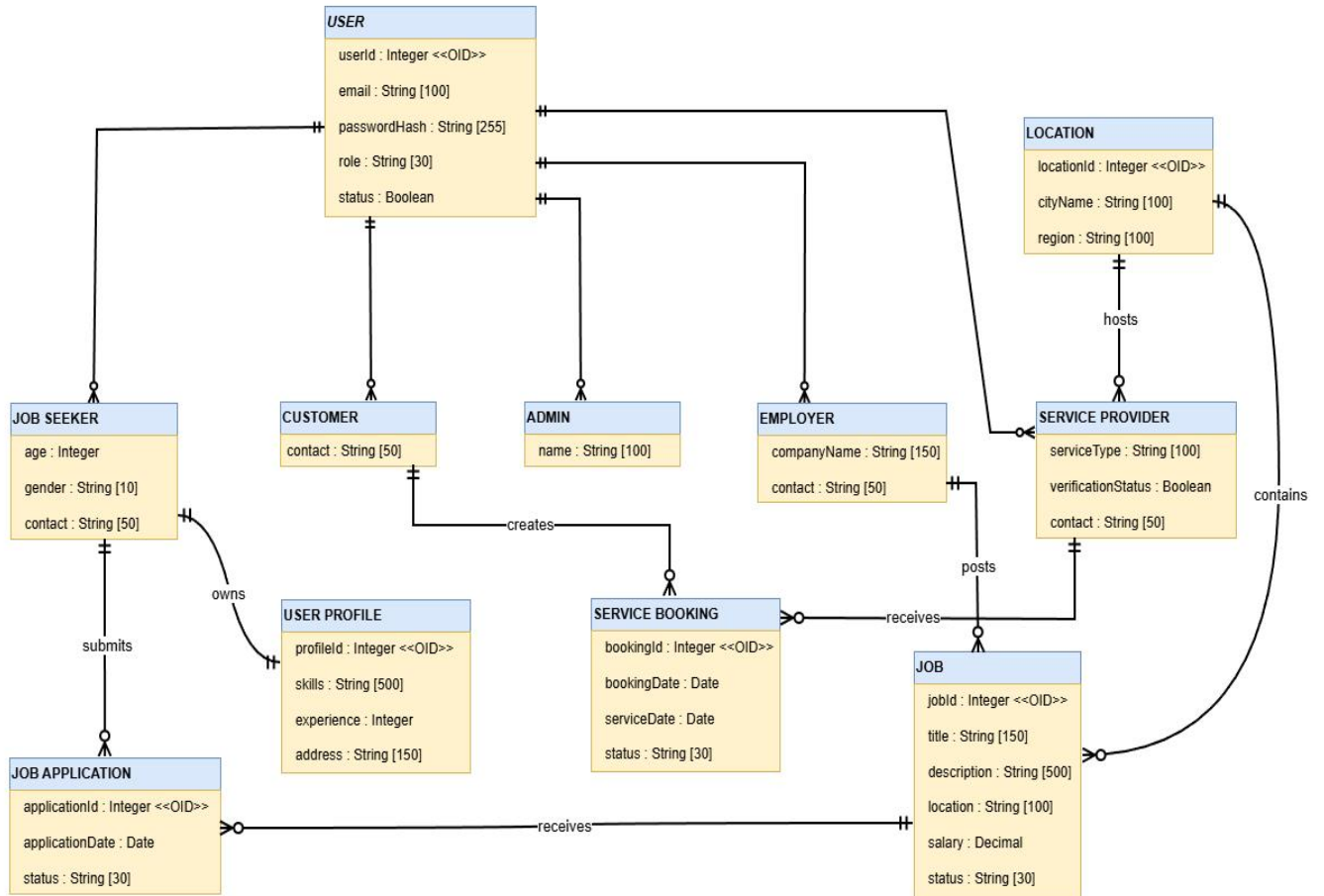


Figure 21: Design of class Model

3.2.2 Persistent model

Shows the database structure of the system, including tables, primary keys, attributes, and relationships used to store users, jobs, bookings, applications, and locations permanently.



Design of Persistent Model

Figure 22: Design Persistent Model

3.3 User Interface Design

S SkillLink Search... Abebe Kebede

Employer

- Dashboard
- Post Job**
- My Jobs
- Candidates
- Interviews
- Messages
- Settings
- Log out

Post a New Job

Fill in the details to attract the best candidates

Job Details

Job Title
e.g., Senior Software Engineer

Job Type
Select type

Experience Level
Select level

Location
e.g., Addis Ababa or Remote

Salary Range (ETB)
e.g., 35,000 - 50,000/month

Figure 23: Employer Post Job Page

S SkillLink Search... Abebe Kebede

Provider

- Dashboard**
- My Services
- Bookings
- Jobs
- Earnings
- Messages
- Reviews
- Settings
- Log out

Welcome back, Dawit!

Here's your business overview for today

[Manage Services](#) [Set Availability](#)

ETB 45,230
Total Earnings
+12.5%

8
Active Bookings
+3

4.9
Average Rating
+0.2

1,234
Profile Views
-5%

Upcoming Bookings [View all](#)

Web Development Project
Abebe Kebede
Dec 15, 2025 10:00 AM
Confirmed

Job Matches [View all](#)

Senior React Developer
TechCorp Ethiopia
95% Match

Figure 23: Service provider Dashboard Page

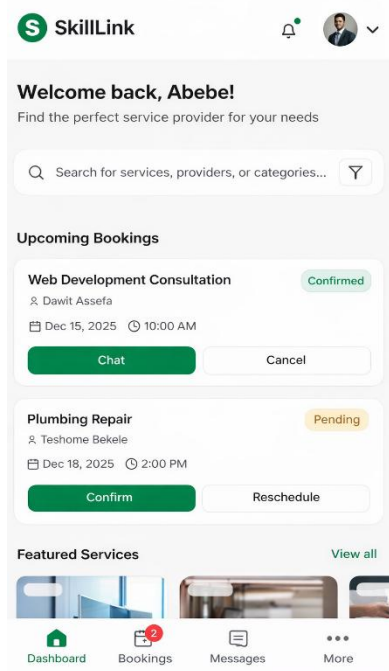
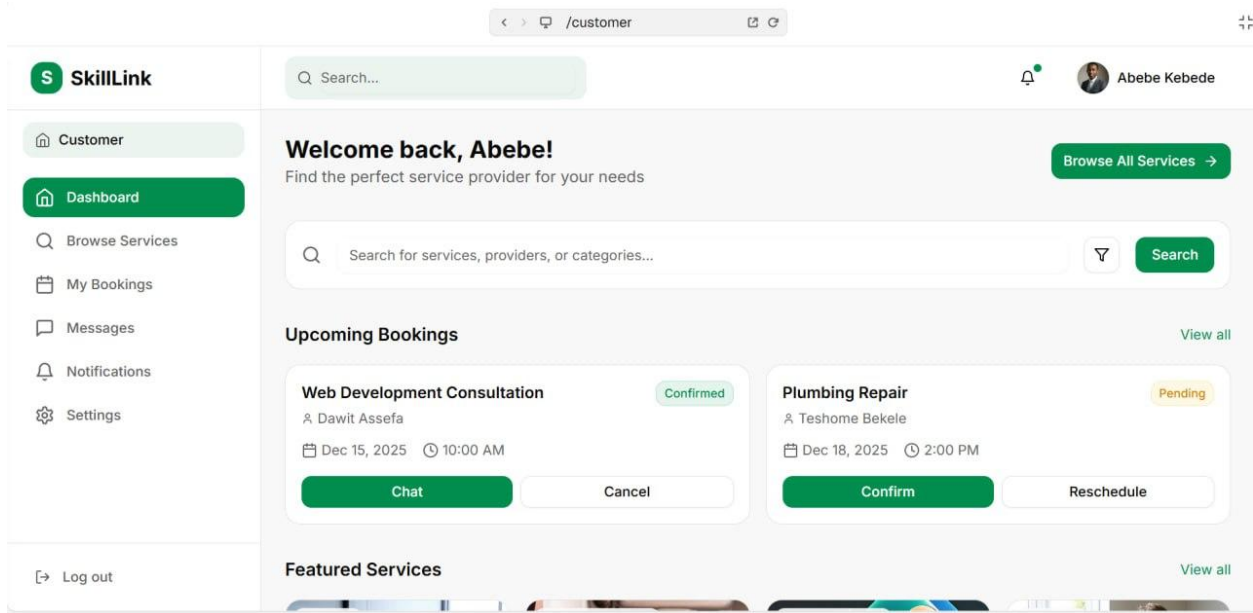


Figure 24: Customer Dashboard Page

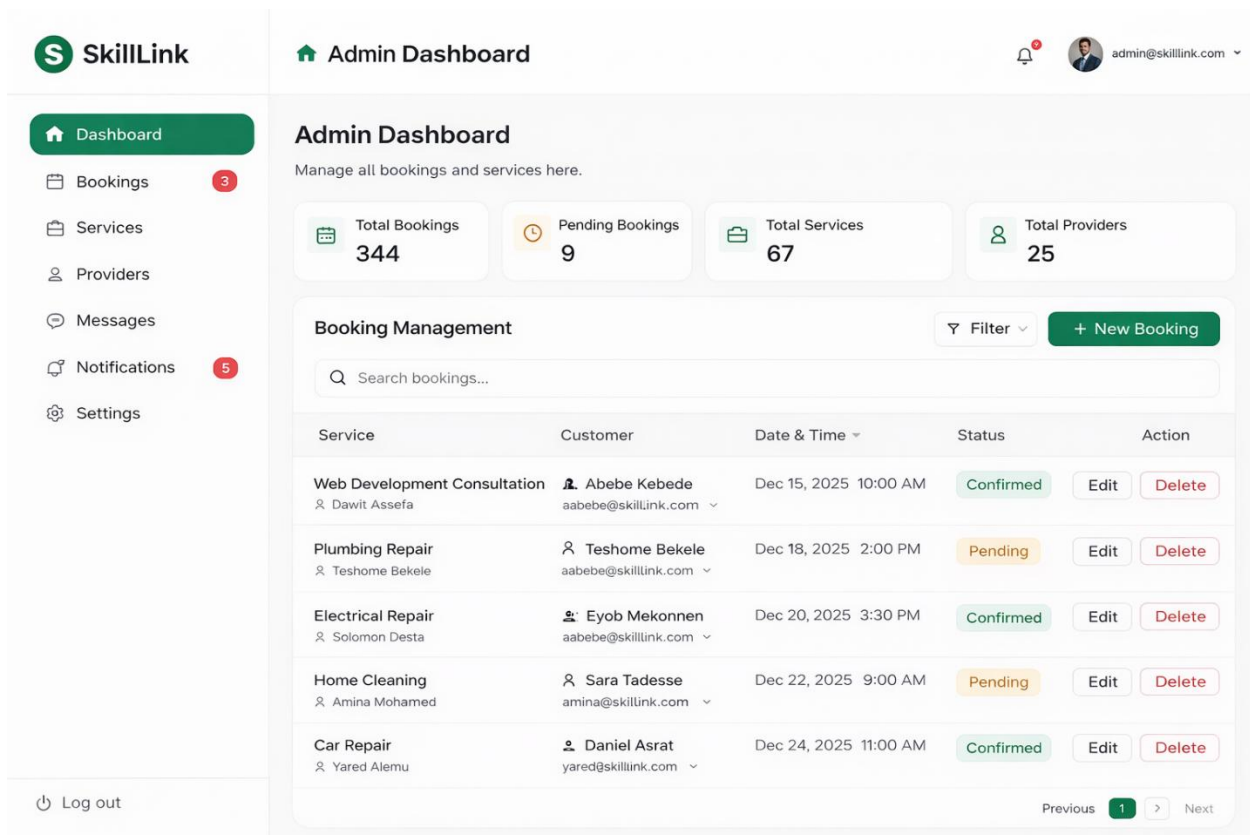


Figure 25: Admin Dashboard

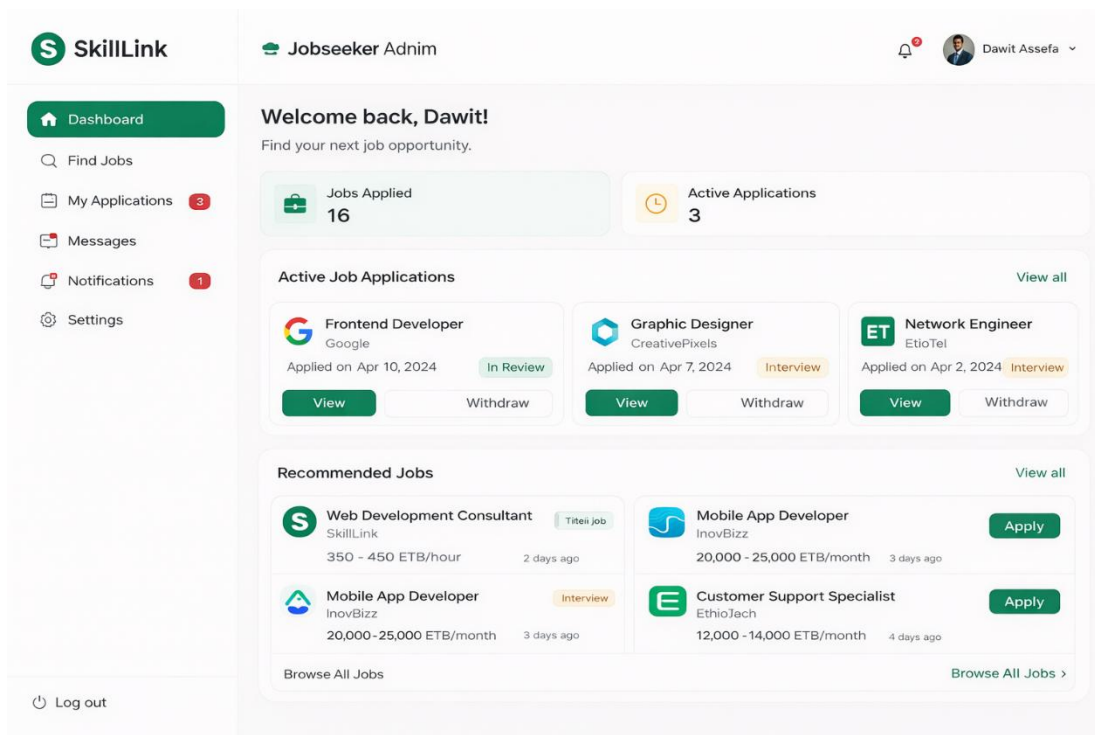


Figure 26: JobSeeker Dashboard

3.1.4 Access Control and Security Modeling

To ensure the integrity and trustworthiness of the **SkillLink** platform, a multi-layered security architecture has been designed. This section details the logical distribution of permissions through Role-Based Access Control (RBAC) and the technical safeguards implemented to protect sensitive user data.

A. Role-Based Access Control (RBAC)

The system utilizes a **Role-Based Access Control (RBAC)** model. This "middle-layer" approach ensures that security is scalable and easy to manage by assigning permissions to roles rather than individual users.

Explanation: users are mapped to specific roles: **Customer, Service Provider, Employer, or Administrator**. Each role is granted specific permissions that define their interaction boundaries. For example, while the "Video Interview" module is accessible to both Customers and Providers to facilitate engagement, "Financial Management" and "Escrow Release" are strictly restricted to authorized roles to prevent fraud.

B. Access Control Matrix

While the RBAC diagram shows the logic, the following **Access Control Matrix** provides the exact technical details of functional permissions. This is essential for preventing unauthorized actions, such as a Service Provider attempting to access the Admin Dashboard.

Feature / Module	Customer	Service Provider	Jobseeker	Employer	Admin
Search & View Profiles	✓	✓	✓	✓	✓
Post Service Listings	X	✓	X	X	✓
Post Job Vacancies	X	X	X	✓	✓
Apply for Jobs	X	X	✓	X	X
Video Interview Access	✓	✓	✓	✓	X
Release Escrow Payment	✓	X	X	X	✓
Delete/Suspend Accounts	X	X	X	X	✓
Access System Analytics	X	X	X	X	✓

Key: (✓) Authorized | (X) Unauthorized

Table 9: Role-Based Access Control (RBAC)

C. Security Specifications (Non-Functional Requirements)

Beyond access control, the following technical "barriers" protect the platform from external attacks and ensure the safety of financial transactions:

- **Authentication:** The system utilizes **JSON Web Tokens (JWT)** for secure, stateless session management, ensuring that user identities are verified across all requests.
- **Data Encryption:** All sensitive data, including user credentials and payment metadata, is encrypted using **AES-256 standards**.
- **Transport Security (TLS):** Communication between the client and server is secured via **HTTPS with TLS 1.2+** to prevent man-in-the-middle attacks.
- **Account Safety & Password Complexity:** To prevent brute-force attacks, accounts lock after **5 failed attempts**. Passwords must be at least **8 characters**, including uppercase, lowercase, numbers, and special characters.
- **Financial Integrity (Escrow System):** The **Escrow system** acts as a security buffer, holding funds securely until both the Customer and Provider confirm task completion, minimizing the risk of financial disputes.

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Appendices

Appendix A: Interview Questionnaire (Multiple Choice)

This questionnaire is designed to identify challenges in the local labor market and trust issues in informal hiring systems.

ይህ ጥያቄ መጠይቅ በአካባቢያዊ የስራ ገበያ ውስጥ የሚገኙ ችግሮችን እና በያልተደነገገ የመቅጠር ሂደት ውስጥ ያሉ የእምነት ችግሮችን ለመለየት ተዘጋጅቷል።

Part I: For Talent / Service Providers

Q1. How do you currently find work or advertise your skills?

እስከሁሉም ክፍሎች እና ስራዎች እንዴት ያስተዋውቃሉ?

- (A) Word of mouth / Personal contacts (በአፍ ንግግር / የግል አገናኝ)
- (B) Social media (Facebook, Telegram, etc.) (ማህበራዊ ሚዲያ (ፌስቡክ፣ ቴሌግራም))
- (C) Agents / Brokers (አገናኞች / ደላላዎች)
- (D) Online platforms (የመስመር ላይ መድረኮች)

Q2. What is the biggest challenge in proving your skills to new clients?

ለአዲስ ደንበኞች ክፍሎችን ለማረጋገጥ የሚያጋጥም ትልቁ ችግር ምንድን ነው?

- (A) Lack of certificates or documents (የማስረጃ እጥረት)
- (B) Lack of trust from clients (የደንበኞች እምነት እጥረት)
- (C) High competition (ከፍተኛ ውድድር)
- (D) No digital profile or portfolio (ዲጂታል ፕሮፋይል እጥረት)

Q3. Have you ever faced a payment dispute after completing a job?

ስራ ካጠናቀቁ በኋላ የክፍያ ክርክር አጋጥሞዎት ያውቃል?

- (A) Yes, frequently (አዎ፣ ብዙ ጊዜ)
- (B) Yes, sometimes (አዎ፣ አንዳንድ ጊዜ)
- (C) Rarely (አልፎ አልፎ)
- (D) Never (አልተፈጠረም)

Q4. Would a verified digital profile help you get better jobs?
የተረጋገጠ ዲጂታል ፕሮፋይል የተሻለ ስራ እንዲያገኙ ይረዳዎታል?

- (A) Strongly agree(በጣም እስማማለሁ)
- (B) Agree(እስማማለሁ)
- (C) Neutral(መካከለኛ)
- (D) Disagree(አልስማማም)

Part II: For Clients / Employers

Q5. How do you usually find workers for urgent services?
አስቸኳይ አገልግሎት ሰራተኞችን እንዴት ትፈልጋላችሁ?

- (A) Friends or neighbors(ጓደኞች / ጎረቤቶች)
- (B) Phone contacts(የስልክ አገናኞች)
- (C) Social media groups(የማህበራዊ ሚዲያ ቡድኖች)
- (D) Online platforms(የመስመር ላይ መድረኮች)

Q6. How do you assess the reliability of a new worker?
አዲስ ሰራተኛ እምነት ያለው መሆኑን እንዴት ታረጋግጣላችሁ?

- (A) Personal recommendation(የሰው ምክር)
- (B) Past work experience(የቀድሞ ስራ ልምድ)
- (C) Trial work(ሙከራ ስራ)
- (D) Ratings and reviews(ደረጃ እና ግምገማ)

Q7. Which feature would increase your payment confidence most?
ከፍተኛ ለመፈፀም የሚያስተምንዎት ባህሪ የትኛው ነው?

- (A) Verified identity badge(የተረጋገጠ መታወቂያ)
- (B) Escrow-based payment(የኤስክሮ ክፍያ)
- (C) Admin dispute resolution(የአስተዳዳሪ መፍትሄ)
- (D) User reviews (የተጠቃሚ ግምገማ)

Q8. Would AI-based matching reduce your hiring stress?

AI መመሪያ ያለው መዛመድ የመቅጠር ጭንቀትዎን ይቀንሳል?

- (A) Yes, very much(አዎ፣ በጣም)
- (B) Yes, somewhat(አዎ፣ በመጠኑ)
- (C) Not sure(አልተረጋገጠም)
- (D) No(አይ)

Appendix B: User Requirements Survey (Multiple Choice)

Q1. User Role

የተጠቃሚ ሚና

- (A) Job Seeker – ስራ ፈላጊ
- (B) Employer – አሰሪ
- (C) Customer – ደንበኛ

Q2. Difficulty of finding verified talent

የተረጋገጠ ባለሙያ ማግኘት ምን ያህል አስቸጋሪ ነው?

- (A) Very difficult(በጣም አስቸጋሪ)
- (B) Difficult(አስቸጋሪ)
- (C) Moderate(መካከለኛ)
- (D) Easy(ቀላል)

Q3. Which features are most important to you? (Select all)

ከሚከተሉት ባህሪዎች አስፈላጊ የሆኑትን ይምረጡ

- ☐ Amharic language support – የአማርኛ ድጋፍ
- ☐ Secure escrow payment – ደህንነታማ ኤስክሮ ክፍያ
- ☐ Video interview – የቪዲዮ ቃለ-መጠይቅ
- ☐ Identity verification – መታወቂያ ማረጋገጫ

Q4. Would you trust an AI-generated match score?

AI የሚያመነጭ የመዛመድ ውጤትን ታምናላችሁ?

- (A) Yes(አዎ)
- (B) No(አይ)

For Online we use Online Questionnaire- Amharic Version

https://docs.google.com/forms/d/e/1FAIpQLSdIIVLafU2ojdZ56VwPZIIIj-XDUSQB-F_JNU_rawaApCSOcA/viewform

Appendix C: Project Plan / Gantt Chart

The SkillLink development follows a two-semester Agile lifecycle to move from conceptualization to a functional MVP.

Phase	Activity Description	Duration	Milestone
I: Analysis	Problem definition, user interviews, and requirement elicitation.	4 Weeks	Requirement Spec
II: Design	Creating Conceptual Class Diagrams, CRC cards, and UI Prototypes.	4 Weeks	Design Document
III: Build	Frontend development with React/Next.js and Backend with Node.js.	8 Weeks	Alpha Prototype
IV: AI Logic	Implementing the AI Matching Engine and location-based filters.	4 Weeks	Smart Matching Live
V: Testing	User Acceptance Testing (UAT) and security auditing.	3 Weeks	Beta Release
VI: Closing	Final documentation, presentation, and deployment to Vercel.	2 Weeks	Project Completion

Appendix D: User Persona Sketches & Journey Maps

To support the UI Prototyping section (2.3.4), the following journey map illustrates the path from search to payment.

The Customer Journey Map:

1. **Entry:** User lands on the Home Screen and selects a service (e.g., "Plumbing").
2. **Search:** The system utilizes the AI Matching Engine to present ranked candidates.
3. **Engagement:** User initiates a Video Interview or messages the provider directly.

4. **Booking:** The user triggers a "Prompt for Payment" via the Escrow system.
5. **Completion:** The provider completes the task, and the admin releases the funds.